Confidence in Inference*

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Abstract

Do people behave as if they know how informative signals are? I axiomatically identify the restrictions on choice implied by certainty about the information structure in a sample environment. Certainty is equivalent to a separability axiom and yields parallel linear indifference curves in the space of samples. This equivalence does not depend on expected utility or Bayesian updating and holds for a broad class of monotone updating and choice rules. A controlled experiment shows that certainty about the information structure is rejected for 95% of subjects. Subjects are insensitive to the stated information structure and instead choose based solely on sample characteristics. Many decisions display a sample-size neglect bias. Using an incentive-compatible confidence elicitation method, I find that sample-size neglect is positively associated with confidence, suggesting that subjects act as if they are uncertain about the information structure even when it is explicitly provided.

keywords: belief updating, confidence, samples of signals, signal uncertainty

JEL codes: C91, D81, D83, D91

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1 Introduction

How do people use ratings and reviews when choosing between products? More broadly, how do decision makers (DMs henceforth) interpret information? A pervasive assumption in theoretical and empirical work is that the mapping from states to signals—the *information structure* (or likelihood)—is known. In information acquisition and design, the DM is assumed to know the experiment she faces; in social learning, she is assumed to know the signal structures observed by others; and in laboratory studies of belief updating, subjects are typically told the information structure explicitly. This paper studies the empirical content of this assumption: what choice behavior is implied by *information certainty*, and do these implications hold?

A large literature relaxes Bayesian updating or expected utility (EU) preferences to accommodate systematic departures from normative benchmarks. The present work asks whether information certainty imposes *testable* restrictions on observable choices *independent* of risk preferences and updating rules. This paper answers this question with an axiomatic identification of the set of choices consistent with information certainty. A controlled experiment then tests whether these restrictions hold.

A simple thought experiment previews the key restriction. Consider Alice, who is choosing between two brands of lightbulbs, A and B, each of binary quality (works or does not work). At time 1, she observes 4 out of 5 positive ratings for A and 1 out of 1 positive rating for B. At time 2, the samples are multiplied one hundred-fold: she observes 400 out of 500 positive ratings for A and 100 out of 100 positive ratings for B. Suppose that the sample size is independent of quality and that, conditional on quality, ratings are independent draws from a *known* information structure. Many readers will find it natural to choose A at time 1—because its evidence is "less extreme" but based on a larger sample—and to switch to B at time 2, when the evidence for B becomes overwhelming. However, this choice pattern is impossible under *any* updating rule and risk preference satisfying mild monotonicity conditions. This impossibility stems from the fact that the DM knows the information structure.

The theoretical framework considers a DM choosing between ex-ante identical objects of binary quality. Each object is accompanied by a sample of signals (e.g., counts of positive and negative ratings). *Information certainty* is formally defined as having degenerate beliefs about the signal distribution conditional on quality. I show that information certainty is equivalent to a behavioral *separability* property under standard regularity axioms. Separability requires that if the DM prefers an object with sample s_1 to one with sample s_2, then she also prefers $s_1 + s_3$ to $s_2 + s_3$ for any additional sample s_3. Theorem 1

establishes that, under mild monotonicity assumptions on updating and preferences, separability and standard regularity axioms are necessary and sufficient for information certainty. However, while this is a general class of models, they rule out the earlier switching choice pattern. Note that under separability, $s_1 \succ s_2$ implies $s_1 + s_1 \succ s_1 + s_2$ and $s_1 + s_2 \succ s_2 + s_2$. Applying transitivity and iterating yields $100 \times s_1 \succ 100 \times s_2$, ruling out the choice reversal of the thought experiment.

Theorem 1 shows further that information certainty implies evaluating samples *linearly* in their signal composition. This implies, geometrically, that indifference curves (ICs henceforth) in the space of samples are parallel lines with constant slope. This captures the tension in the thought experiment. When the likelihood is known, each signal has a fixed interpretation, so the marginal tradeoff between signal types does not depend on the observed signals. To sum up, Theorem 1 identifies the set of choices consistent with information certainty and provides a comprehensive choice-based falsification test based on the shape of ICs.

What kind of model could rationalize the thought experiment? If the DM faces uncertainty about the information structure, then she can display the switching pattern of the thought experiment. In particular, additional observations can change how earlier signals are interpreted. A single positive rating for B may be dismissed as noise, whereas 100 positive ratings are not. In terms of ICs, this corresponds to a *gradual fan-out*: the slope, which captures the trade-off between signal types, changes with the realized sample depending on its sample distribution and size.

Are there other plausible explanations beyond uncertainty about the information structure? Introspection suggests that, even when the information structure is explicitly provided, DMs may employ various heuristics based on sample characteristics. Two natural patterns, suggested by the thought experiment, are common. Firstly, people appear to rely primarily on the *sample proportion* of favorable signals, neglecting the sample size. This pattern, *sample-size neglect*, has been documented in belief updating and is more frequent when samples are large. Secondly, people are typically more confident in decisions based on large samples. Are these patterns compatible or contradictory with information uncertainty? Proposition 1 shows that information uncertainty rationalizes both of these patterns. Under a mild monotonicity condition on subjective uncertainty, when samples are sufficiently large, it is optimal to choose the option with the higher proportion of favorable signals. Thus, a DM may rationally focus on proportions and simultaneously report higher confidence *when samples are large*. Therefore, information uncertainty predicts a positive correlation between confidence and sample-size neglect: neglect arises precisely when the DM is confident that sample sizes are "large enough" to be neglected.

Several motivations jointly compel an experimental investigation. First, the thought experiment suggests a natural violation of information certainty, but it is neither a systematic test nor is it known whether such violations occur empirically. Second, while previous works have studied belief updating with samples, little is empirically known about *choice behavior* in this environment. Lastly, Theorem 1 requires objects to be of binary quality as well as the sample size to be independent of the quality. These are restrictions that are hard to guarantee in the real world, necessitating a controlled experiment. Further, Proposition 1 conjectures a relationship between choices and confidence in choice. Measuring this latter variable cleanly also requires a controlled experiment. Therefore, a controlled experiment allows for 1) a complete test of information certainty, 2) a more extensive empirical documentation of choices between objects with samples, and 3) allows for the choice scenario to tightly mirror the theoretical framework.

I conduct an experiment in which subjects choose between ex ante identical binary-quality objects, conditional on observed samples of binary signals. The standard experimental environment of belief updating mirrors my theoretical environment. Subjects are asked to make 10 incentivized choices between objects associated with samples of signals of sizes 4, 10, and 25. They additionally completed six belief-elicitation tasks for robustness. As in standard belief-updating experiments, subjects are given the information structure explicitly and told about the correlation between signals and object quality. I also implemented three between-subject treatments that varied the explicitly stated information structure. After each choice, subjects reported their confidence using a novel incentive-compatible mechanism. The experiment is preregistered and designed to answer three questions.

1. Are choices consistent with information certainty? No. By Theorem 1, information certainty is equivalent to separability, which in this sample environment implies linear evaluation of samples and therefore parallel ICs in the sample space. I find that 95% of subjects exhibit systematic violations of this restriction. These violations are also large in magnitude and cannot be rationalized by noise. These violations are robust and even stronger in a preregistered attentive subsample defined by monotonicity compliance. The subsample and other tests show that these results are not driven by noise, inattention, or randomization. This occurs despite being explicitly told about the information structure. Moreover, ICs recovered from belief-elicitation data are not parallel, suggesting that the violations are not driven by the choice-task format.

2. Do choices respond to the stated information structure? No. If subjects condition on the stated information structure, then changing the information structure across treatments should shift their ICs. The treatments vary in the stated information structure and whether it is known with certainty. Despite these theoretically significant differences, ICs are nearly identical across the three treatments, both in comparative choices and in belief-elicitation tasks. Consistent with this invariance, many choices display sample-size neglect: holding the proportion of favorable signals fixed, subjects behave as if increasing sample size carries little weight. This suggests that subjects mostly use the sample characteristics and do not incorporate the information structure's parameters in their choice.

3. Are the observed patterns consistent with uncertainty about the information structure? Yes. Choice data alone cannot prove that subjects must be uncertain about the information structure. However, I document a robust pattern predicted by models of information-structure uncertainty. Proposition 1 shows that sample size-neglect arises precisely when subjects are confident that the samples are "large enough". I find choice confidence is strongly positively related to sample-size neglect. If sample-size neglect was driven by heuristic use arising from the complexity or inattention concerns, then we would expect a negative correlation instead.

Taken together, the results reject the ubiquitous information certainty benchmark in a common sample-based choice environment. More broadly, they suggest that information-structure uncertainty may matter even in settings where the information structure is objectively known and explicitly communicated, such as laboratory studies of belief updating. This is especially important because many experimental works attribute various deviations to non-Bayesian updating or non-EU preferences. But if information certainty fails, when the information structure is explicitly given, then those inferences are not identified. The main contribution of this paper is to provide a general, choice-based test of information certainty—a restriction that is independent from updating rules and risk preferences—and to show experimentally that this restriction fails in an empirically relevant and classic experimental paradigm.

Literature. A growing experimental literature studies belief updating under ambiguous or uncertain information and evaluates specific models of ambiguity and updating (e.g., Shishkin and Ortoleva, 2023; Liang, 2023; Epstein and Halevy, 2024; Aina and Schneider, 2026). Other work appeals to uncertainty about information or updating to explain observed behavioral biases, including the law of small numbers and systematic over- and

under-inference (e.g., Benjamin et al., 2016; Augenblick et al., 2025; Enke and Graeber, 2023).

This paper fills several gaps in the literature. First, differences in elicited beliefs between subjects given explicitly certain and uncertain information structures do not identify whether subjects do indeed *subjectively* perceive information as certain or uncertain. Second, belief elicitation may be noisy or distortionary (see Danz et al. (2022)). Third, no established belief elicitation method exists under ambiguity and preferences sensitive to second-order uncertainty. Both of which are relevant channels in the context of information uncertainty. Finally, the literature has mainly tested specific models of certainty and uncertainty in information via their implications. By contrast, Theorem 1 offers an *exact identification* of the *choice behavior* of *every model* of information certainty. This allows for a revealed-preference falsification test of this class of models without requiring parametric assumptions or belief elicitation.

The experimental findings also complement previous works. This paper finds that information uncertainty is likely a real driver of choice, supporting previous work that appeals to it as an explanation for observed behavioral biases. Further, the results suggest that information uncertainty may be behaviorally relevant even when the information structure is explicitly stated, a common feature of laboratory belief-updating designs. Second, as in Griffin and Tversky (1992), I also find the subject overweight sample proportion to size. But crucially, the thought experiment, theory, and experimental results suggest this overweighting is not constant and occurs gradually as the sample size increases. Finally, this work also contributes to the discussion of updating under ambiguity, as neither Prior-by-Prior nor Maximum Likelihood Updating, the two most common updating rules under Maxmin EU, can accommodate the thought experiment.

My results additionally suggest that the link between *objective* and *subjective* information structures is likely much looser than previously thought. This has important implications in light of Shmaya and Yariv (2016), who show that Bayesian updating is difficult to reject without additional assumptions on the DM's *subjective* information structure. The current paper reinforces the flip side of this coin: under standard benchmark assumptions—i.i.d. signals with known likelihoods—*every* monotonic updating model is rejected by almost all subjects' choices. Taken together, these results suggest that, alongside the extensive focus on updating rules, there is a need to better understand the subjective "mental model" of the information structure to explain choice behavior.

The current work also designs and implements a novel confidence elicitation mechanism. Existing approaches to eliciting confidence often fail to be incentive-compatible, rely on strong behavioral assumptions, or require restrictive environments (e.g., Coff-

man, 2014; Karni, 2018, 2020; Chambers and Lambert, 2021; Halevy et al., 2023; Enke and Graeber, 2023; Nielsen and Rigotti, 2023). I propose a simple mechanism that can be appended to standard choice tasks. The mechanism is easy for subjects to understand, inexpensive to implement, and incentive-compatible for a broad class of theories.[1] It yields a confidence measure that admits meaningful interpersonal comparisons and correlates strongly with standard (unincentivized) self-reports, providing a practical tool for studying second-order uncertainty in information processing and other contexts.

2 Theory

A DM chooses between ex-ante identical binary-valued objects, good or bad. Each object has a probability p of being good and is associated with a sample of simultaneously observed signals. A sample of signals is a K-dimensional vector whose entries are natural numbers. Samples are denoted as $s = (s^1, .., s^K) \in \mathbb{N}_0^K$, where s^k denotes the number of signals of type k. For example, each object could be a lightbulb, and a sample would be a set of different types of ratings. One common example is star ratings, which feature ratings from 1 to 5 stars. As objects are ex-ante identical, the choice depends only on their samples. The primitive of interest is a weak order $\succsim \subseteq \mathbb{N}_0^K \times \mathbb{N}_0^K$.

Denote by $\sigma_g, \sigma_b \in (0,1)^K$ the likelihood vectors over the various types of signal realization conditional on the good and bad states, respectively. Hence, $\sigma_{g,k}$ denotes the probability of a signal realization being of type k conditional on the object being good. In this choice environment, the following general class of representations will be the object of study.

Definition 1. *A preference relation $\succsim$ is rationalizable by information certainty if there exist likelihood vectors $\sigma_g = (\sigma_{g,1}, ..., \sigma_{g,K})$ and $\sigma_b = (\sigma_{b,1}, ..., \sigma_{b,K})$ such that $\forall s_1, s_2$,*

$$s_1 \succsim s_2 \;\Leftrightarrow\; \frac{\prod_{k=1}^K \sigma_{g,k}^{s_1^k}}{\prod_{k=1}^K \sigma_{b,k}^{s_1^k}} \geq \frac{\prod_{k=1}^K \sigma_{g,k}^{s_2^k}}{\prod_{k=1}^K \sigma_{b,k}^{s_2^k}}.$$

Choices are rationalizable by information certainty if there exists some known information structure such that the object chosen has a higher sample likelihood ratio (LR) of being good. The overwhelming majority of updating rules, including Bayesian and

[1] Pęski and Stewart (2025) further shows theoretically that the method is non-distortionary and incentive-compatible for an EU DM. In other words, the confidence elicitation mechanism does not incentivize subjects to change their initial report. They show that, in general, very few mechanisms satisfy this property even for a Bayesian EU DM.

Grether updating, are increasing in the LR.[2] I assume only that the information structure is such that for all k, $\sigma_{g,k} \in (0,1)$ and $\sigma_{b,k} \in (0,1)$, a full support condition. Hence, this allows for incoherent information structures in which the signal realization probabilities need not sum to 1. Additionally, in terms of choice, the representation imposes only that the DM prefers a higher probability of a good object, which encompasses virtually all theories of choice under risk; some examples include EU, rank-dependent EU (Quiggin, 1982), cautious EU (Cerreia-Vioglio et al., 2015), and others. Hence, this is a very general representation with only mild requirements on the rationality of updating and choice, with the key assumption being instead that there is no uncertainty on the information structure.

The reader may be curious about the assumptions of binary values and signal realizations being i.i.d. These are taken up in detail in the discussion section. I show that the binary-value assumption is necessary for identification, while the iid assumption is implied by information certainty and the fact that signal orders are not observed via an exchangeability argument.

Despite its generality, this representation turns out not to be silent on choices. Consider the thought experiment from the introduction again: Alice is choosing between two brands of lightbulbs, A and B. Let ratings have respective likelihoods $\sigma_{g,A}$ and $\sigma_{g,B}$ of being positive conditional on the lightbulb working. Denote by $\sigma_{b,A}$ and $\sigma_{b,B}$ the likelihood of a positive rating conditional on the light bulb not working. Suppose, for the sake of illustration, that Alice is Bayesian and EU maximizing. Then the first choice implies that the posterior probability of A working is higher than that of B working. And because quality is binary, this implies the posterior ratios must satisfy

$$\frac{\Pr(\text{A works} \mid \text{4 out of 5 positive})}{\Pr(\text{A fails} \mid \text{4 out of 5 positive})} > \frac{\Pr(\text{B works} \mid \text{1 out of 1 positive})}{\Pr(\text{B fails} \mid \text{1 out of 1 positive})}. \tag{1}$$

Applying the Bayesian updating formula, and canceling out the priors, which are identical for A and B, and rewriting in terms of signal likelihoods, given the iid assumption, gives

$$\frac{{\sigma_{g,A}}^4(1-\sigma_{g,A})}{{\sigma_{b,A}}^4(1-\sigma_{b,A})} > \frac{\sigma_{g,B}}{\sigma_{b,B}}. \tag{2}$$

[2] Chan (2025) shows that monotonicity with respect to the LR is satisfied experimentally and is, in general, one of the most robust properties of belief updating.

And by a similar calculation, if Alice chooses to pick B over A at time 2, then

$$\frac{{\sigma_{g,A}}^{400}(1-\sigma_{g,A})^{100}}{{\sigma_{b,A}}^{400}(1-\sigma_{b,A})^{100}} < \frac{{\sigma_{g,B}}^{100}}{{\sigma_{b,B}}^{100}}. \tag{3}$$

Notably, the inequality at time 2 is simply the inequality at time 1 raised to the power of 100. Hence, this behavior cannot be rationalized by a Bayesian for any information structure, as long as it is known with certainty. It is also evident that monotonic updating with respect to the sample LR, as well as preference for a higher probability of a working lightbulb, suffices to rule out switching.

The thought experiment offers one falsification test of information certainty. But it does not provide an exact test of whether DMs choose *as if* they had information certainty. Instead, a complete identification of choices consistent with Definition 1 is required. I show that two axioms, alongside completeness and transitivity, are necessary and sufficient for the representation.

Axiom 1 (Separability). *For all s_1, s_2, s_3, $s_1 \succsim s_2$ if and only if $s_1 + s_3 \succsim s_2 + s_3$.*

Axiom 2 (Archimedean). *For all $s_1 \succ s_2 \succ s_3$, there exists $N = n_1 + n_2 = m_1 + m_2$ with $n_2, m_1 > 0$ such that $n_1 s_1 + n_2 s_3 \succ N s_2 \succ m_1 s_1 + m_2 s_3$.*

Separability links a DM's preference over samples to the marginal effect of additional samples. In particular, if an object with sample s_1 is chosen over another object with s_2, then for any sample s_3, the DM prefers an object with sample $s_1 + s_3$ to one with $s_2 + s_3$. As the introduction argues, separability encapsulates the intuition that information certainty implies an unchanging interpretation of signals when receiving additional observations. The Archimedean requirement implies that no sample is infinitely more informative than another, and rules out fully revealing signals.

Theorem 1 shows that a DM with complete, transitive, separable, and Archimedean preferences chooses as if she is certain about the information structure and updates her beliefs monotonically with respect to the sample LR.

Theorem 1 (Certainty in Information Characterization).
The following are equivalent:

1. *The relation $\succsim$ is rationalizable by information certainty.*

2. *The relation $\succsim$ is complete, transitive, Archimedean, and separable.*

3. *There exists a vector* $u = (u_k)_{k=1}^K \in \mathbb{R}^K$*, and for all* s_1 *and* s_2

$$s_1 \succsim s_2 \Leftrightarrow u \cdot s_1 \geq u \cdot s_2.$$

Proof Sketch: Given that rationalizability by information certainty is to evaluate samples by their LRs, it is immediate that the log of the LR has a linear representation. The direction of $3 \Leftrightarrow 1$ is proven in this manner. Given the linear representation of 3, a decision-theoretic argument shows that the "expected utility" representation over samples is equivalent to the axioms, yielding $3 \Leftrightarrow 2$.

Theorem 1 shows that a wide class of models, in conjunction with information certainty, imply choice behaviors with a linear utility representation in this sample environment. Graphically, if the signals are binary-valued, then the ICs are parallel lines. Consider Figure 1, where the x-axis and y-axis denote the number of bad and good signals, respectively. Figure 1 features samples of size 4, 10, and 20, which are exactly those studied in this paper's experimental part. Therefore, any sample is a point on the plane, and the experimental setting permits recovery of ICs. Representations falling under Theorem 1 predict parallel and linear ICs. However, the thought experiment requires that there must exist samples s_1 and s_2 such that $s_1 \succ s_2$ while $k \times s_1 \prec k \times s_2$. This implies naturally that ICs cannot be parallel lines.

If information certainty is violated, what is the right descriptive model? Table 11 shows that essentially all updating rules of the literature are rejected by the thought experiment.

Table 1: Updating Rules Rejected by Thought Experiment and Relation to Theorem 1

Updating Rule[a]	Theorem 1	Indifference Curves
Bayesian updating (Bayes and Price, 1763)	Yes	Parallel Lines
Grether updating (Grether, 1980; Möbius et al., 2022)	Yes	Parallel Lines
Weighted Bayesian (Epstein et al., 2010; Kovach, 2021)	Yes	Parallel Lines
Divisible updating (Cripps, 2018)	Yes	Parallel Lines
Confirmatory bias (Rabin and Schrag, 1999)	Yes	Parallel Lines
Prior-by-prior updating (Pires, 2002)	No	Homothetic[b] Lines
Max. Lik. updating (Gilboa and Schmeidler, 1993)	No	Homothetic Lines
Size/proportion model (Griffin and Tversky, 1992)	No	Homothetic Curves
Coarse Bayesian (Jakobsen, 2023)	No	Thick Parallel Lines

[a] The online appendix goes over each of these rules in detail. The only updating rule, with information certainty, that can accommodate the thought experiment to the author's knowledge, must relax monotonicity with respect to the LR, see Dominiak et al. (2023).
[b] ICs are "homothetic" if the slope is constant anywhere on a given ray through the origin, i.e., constant under scalar multiplication.

It turns out that the thought experiment places additional restrictions beyond those characterized by models of information certainty. The thought experiment requires $s_1 \succ s_2$ while $k \times s_1 \prec k \times s_2$, and the ICs of Figure 1 makes it clear. Slopes of samples of *same* proportion must change as *sample size increases*. This is exactly the intuition of the thought experiment, which highlights that inference from a sample isn't just based on proportion but also on the size. Geometrically, Figure 1's panel A, the slopes of the ICs through the samples $(0, 4)$ and $2.5 \times (0, 4) = (0, 10)$ differ via non-parallelism. In Figure 1's panel B, consistent with the thought experiment, the slopes of the ICs through the samples $(2, 8)$ and $2.5 \times (2, 8) = (5, 20)$ differ via non-linearity. In addition, ICs consistent with the thought experiment must *gradually fan-out*. Fanning out is important as it implies the DM switches to the object with a higher proportion of good signals.

Note that prior-by-prior, maximum likelihood updating, and the size-proportion model allow for slopes to change between samples of *different* proportions but not between samples of the same proportion, even when they differ in size. In prior-by-prior and maximum likelihood updating, the DM has uncertainty about the sample interpretation and evaluates according to the worst and most likely interpretations, respectively. However, only the sample proportion is used to infer which interpretation is the worst and most likely. Therefore, the interpretation is independent of sample size, and the slope of indifference curves does not change with size. In the size-and-proportion model, the DM assigns weights to both proportion and size, but these weights are constant and do not interact with each other.

Accommodating the Thought Experiment. A model that accommodates the thought experiment, therefore, needs signal interpretation to change in both sample size and ratio. The standard Bayesian EU DM with information uncertainty can, in fact, accomplish this.

Example. Suppose conditional on quality $q \in \{g, b\}$ the likelihoods for both ratings of lightbulb A and B are randomly determined by $\sigma_g \sim U[0.5, 1]$ and $\sigma_b \sim U[0.3, 0.8]$. Signal realizations are then drawn iid given σ's value. This could be caused by the customer being unsure about how representative their electric wiring is. That is, they believe that if the lightbulbs work for their house, then ratings have at least a 50% chance of being positive. However, if the lightbulbs do not work at their house, then they believe potentially 80% could still give it positive ratings. Then, the LRs for the two decisions display the desired switching patterns.

A Bayesian DM who updates on the uncertainty in the information structure naturally

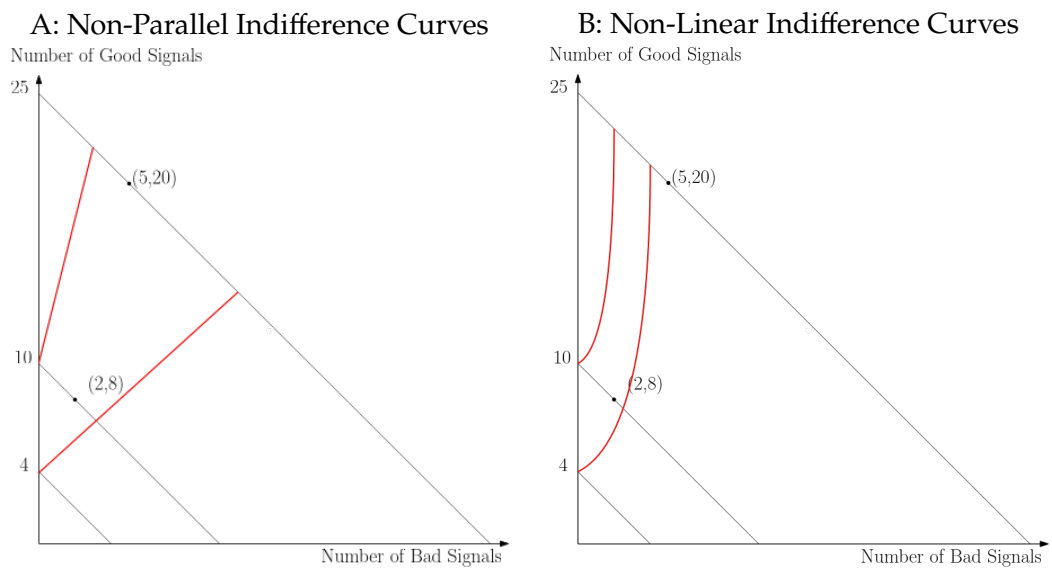


Figure 1: Thought Experiment Consistent Indifference Curves

Notes: The x-axis denotes the number of bad signals. The y-axis denotes the number of good signals. In red are two ICs consistent with the thought experiment.

becomes more sensitive to the sample proportion as the sample size grows, because she is learning about that uncertainty. This is mirrored in the thought experiment: 1 out of 1 positive ratings may be a fluke, but surely 100 out of 100 positive ratings must be reliable.

Although a Bayesian EU DM facing uncertain information and updating this uncertainty as sample sizes grow can generate the switching pattern, one might still argue that real-world DMs are unlikely to engage in such complex updating. Furthermore, they may not even have well-formed second-order beliefs about likelihoods. Instead, they may rely on basic heuristics. In the thought experiment, it is certainly plausible that DMs choose solely on the basis of sample characteristics, without even entertaining information structures cognitively.

The following result shows that these perspectives can be reconciled as heuristic use can arise as an optimal response to uncertainty in information. I also show that heuristic use due to information uncertainty has testable implications.

Suppose signals are binary. Denote by $\Sigma = (0,1)^2$ the set of possible information structures. And let $F \in \Delta(\Sigma)$ be a probability measure over information structures. We assume that F has marginals F_g and F_b that admit decompositions into finitely many atoms and absolutely continuous parts whose densities are continuous and strictly positive on finitely many intervals contained in $(0,1)$.[3] Let $\succsim_F$ be the choice of a Bayesian EU DM

[3]Finiteness is important to rule out pathological cases which permit infinitely oscillating posteriors and

who believes the information structure of each object is drawn from F independently and chooses between objects by updating on the quality and the information structure given the sample. If a Bayesian with belief F, when facing a choice between samples of the same size, always chooses the sample with a higher proportion of good signals, then F is called *monotone*. Hence, uncertainty about the information structure is monotone if good signals are unambiguously better than bad signals. Let $\succsim_h$ be the choice of a DM who always picks the object with the sample with a higher ratio of good signals. The following proposition characterizes the optimality of such a rule.

Denote by $V(s_1, s_2| \succsim, F)$ the expected value from a choice between objects with samples s_1 and s_2, given a choice rule $\succsim$ and information uncertainty F.

Proposition 1 (Asymptotic Optimality of Sample Size Neglect)**.**
$\forall F$ monotone, and $\forall$ s_1 and s_2, the following holds

$$\lim_{n\to\infty} \left[V(ns_1, ns_2| \succsim_F, F) - V(ns_1, ns_2| \succsim_h, F)\right] = 0,$$

The proof is relegated to Appendix A.2.

Consider a DM who knows nothing about the information structure except that good signals are weakly better than bad signals. The proposition says that she can do as well as any Bayesian asymptotically by neglecting the sample size. Hence, this proposition supports the intuition that confidence increases in the second choice of the thought experiment when choosing between larger samples. It establishes a link between confidence and the tendency to neglect sample size. When the DM believes they have observed a sufficiently large sample, they neglect its size. The prediction is then that the DM is more confident in her choices that neglect the sample size.

3 Experimental Design

3.1 Goals.

In order to obtain a clean revealed preference test of information certainty, the experimental setting was necessary. Theorem 1's characterization requires that object quality to be unrelated to sample size; furthermore, binary quality is a necessary condition. Therefore, the main portion of the experiment focuses on studying choice behavior in an environment that exactly mirrors the theoretical framework. By recovering choices while

guarantee convergence is well-behaved.

varying sample size and proportion, I can identify the ICs and directly test information certainty. Additionally, I vary the stated information structure across subjects to test whether behavior responds to likelihoods or instead primarily tracks sample statistics (e.g., proportions). Finally, motivated by Proposition 1, the design elicits a choice-by-choice measure of confidence to test whether confidence comoves with sample-size neglect.

3.2 Environment

Subjects were told there are two types of boxes. A *golden* box is "good" and a *wooden* box is "bad," with prior probability $1/2$ each. Each box contains 10 balls, red or blue, and the composition depends on the box type. Subjects never observe the type directly. Instead, they observe a sample of balls drawn *with replacement* from each box. Red balls are always described as "good" signals: holding sample size fixed, a sample with more red balls is always more favorable evidence that the box is golden. Hence, the proportion of red balls in boxes, which differed by type, denotes the information structure. The treatment section elaborates on the composition. A comprehension quiz ensured subjects understood (i) sampling with replacement and (ii) this monotonicity property.

This environment matches the theoretical primitives: binary states (golden vs. wooden), binary signals (red vs. blue), and conditionally i.i.d. sampling within a box given its type.

3.3 Task overview and timing

Subjects are randomly assigned to one of three treatments and complete three task blocks in random order: two *comparative choice* blocks and one *belief updating* block. The comparative choice block features 10 choices between boxes that are each associated with a sample of signals. Sample sizes are varied between the two blocks to identify and recover ICs. The belief updating block features 6 belief elicitation tasks about the posterior of boxes associated with a sample of signals. Both tasks permit the recovery of ICs. The comparative choice blocks are faithful to the revealed-preference approach and mirror the theoretical objects. The belief-updating block follows a standard belief-elicitation approach for robustness. After each choice, subjects completed a confidence elicitation task. Figure 2 summarizes the experimental flow.

At the end of the experiment, one decision was randomly selected for payment.[4] Additionally, depending on the subject's choice in the confidence elicitation task for this decision, detailed next, they may see additional information and revise their choice.

[4]Subjects are told that one decision was already randomly selected for payment before the start, but would only be revealed to them at the end. This theoretically rules out ambiguity and hedging motives.

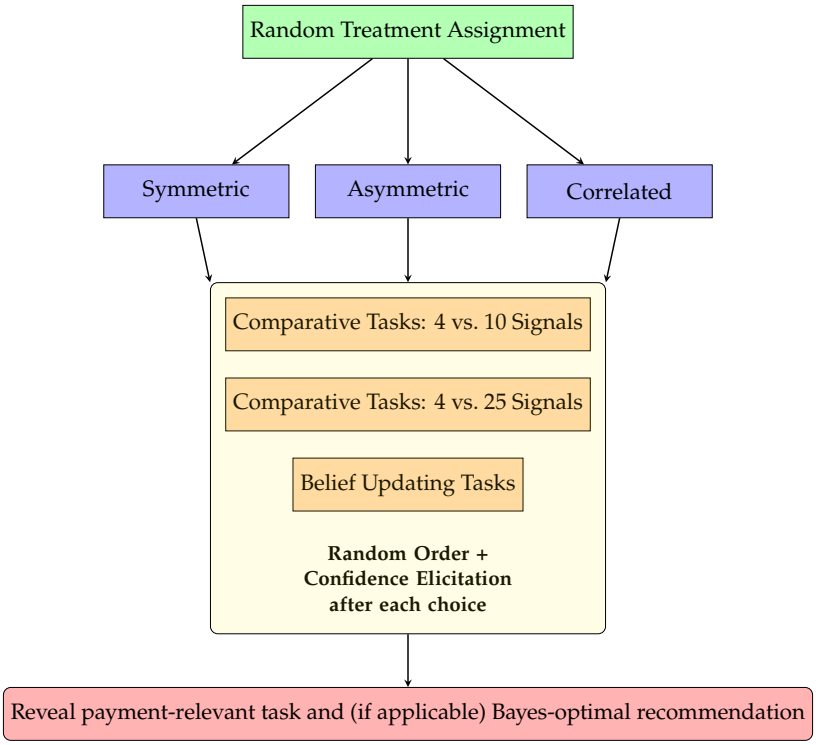


Figure 2: Main experimental parts and timing

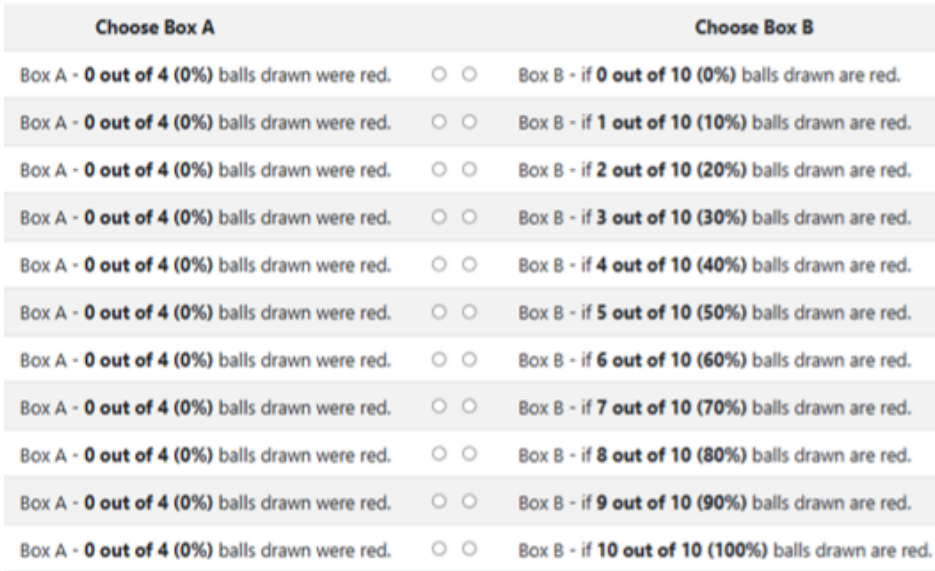
3.4 Experimental Tasks

Throughout the tasks, the subject faces a multiple-price list type interface as illustrated in Figure 3. They are given two boxes, A and B, and may earn a bonus payment if the chosen box is golden. In the comparative choice task, as per Figure 3, Box A has already drawn a sample of signals. In the belief updating task, they are told Box A is golden with a fixed probability p. They know, for both tasks, that Box B, has yet to draw a sample. The interface prompts them to select a "switching point" on a choice list that reveals the number of red balls Box B needs to draw to be weakly preferred to Box A.[5] Hence, it enables the recovery of ICs. For example, Figure 3's choice recovers the minimal number of red balls out of 10 required for Box B to be considered more likely to be golden than Box A, which has drawn 0 out of 4 red balls. In the belief updating task, Box A has a fixed exogenous probability p to be golden instead of having a sample of signals.

There are several variations in the choices. These changed the number of red balls Box A drew, the total number of balls that Box B drew (in comparative choice), and the probability p (in belief updating). Table 2 goes over these in detail. In the comparative task, Box A always drew 4 balls, and could have 0 to 4 red balls total. In the belief updating

[5]The MPL list enforces single switching. Subjects need only click once on their preferred switching point; the interface automatically fills out other choices in a consistent manner.

task, p is either 0.25 or 0.75. Finally, Box B drew either 10 or 25 balls total in both tasks.



Choose Box A		Choose Box B
Box A - **0 out of 4 (0%)** balls drawn were red.	○ ○	Box B - if **0 out of 10 (0%)** balls drawn are red.
Box A - **0 out of 4 (0%)** balls drawn were red.	○ ○	Box B - if **1 out of 10 (10%)** balls drawn are red.
Box A - **0 out of 4 (0%)** balls drawn were red.	○ ○	Box B - if **2 out of 10 (20%)** balls drawn are red.
Box A - **0 out of 4 (0%)** balls drawn were red.	○ ○	Box B - if **3 out of 10 (30%)** balls drawn are red.
Box A - **0 out of 4 (0%)** balls drawn were red.	○ ○	Box B - if **4 out of 10 (40%)** balls drawn are red.
Box A - **0 out of 4 (0%)** balls drawn were red.	○ ○	Box B - if **5 out of 10 (50%)** balls drawn are red.
Box A - **0 out of 4 (0%)** balls drawn were red.	○ ○	Box B - if **6 out of 10 (60%)** balls drawn are red.
Box A - **0 out of 4 (0%)** balls drawn were red.	○ ○	Box B - if **7 out of 10 (70%)** balls drawn are red.
Box A - **0 out of 4 (0%)** balls drawn were red.	○ ○	Box B - if **8 out of 10 (80%)** balls drawn are red.
Box A - **0 out of 4 (0%)** balls drawn were red.	○ ○	Box B - if **9 out of 10 (90%)** balls drawn are red.
Box A - **0 out of 4 (0%)** balls drawn were red.	○ ○	Box B - if **10 out of 10 (100%)** balls drawn are red.

Figure 3: MPL choice elicitation interface (comparative 4 vs. 10)

Notes: Single crossing is enforced. Subjects need only choose on one row, and single crossing consistent choices are automatically selected. Subjects are taught explicitly about how to use the interface.

Table 2: Summary of elicited thresholds

Comparative: 4 vs. 10	**Comparative: 4 vs. 25**	**Belief-elicitation**
$A: (4,0)$ vs. $B: (10-x,x)$	$A: (4,0)$ vs. $B: (25-x,x)$	$A: \ell_{0.25}$ vs. $B: (4-x,x)$
$A: (3,1)$ vs. $B: (10-x,x)$	$A: (3,1)$ vs. $B: (25-x,x)$	$A: \ell_{0.25}$ vs. $B: (10-x,x)$
$A: (2,2)$ vs. $B: (10-x,x)$	$A: (2,2)$ vs. $B: (25-x,x)$	$A: \ell_{0.25}$ vs. $B: (25-x,x)$
$A: (1,3)$ vs. $B: (10-x,x)$	$A: (1,3)$ vs. $B: (25-x,x)$	$A: \ell_{0.75}$ vs. $B: (4-x,x)$
$A: (0,4)$ vs. $B: (10-x,x)$	$A: (0,4)$ vs. $B: (25-x,x)$	$A: \ell_{0.75}$ vs. $B: (10-x,x)$
		$A: \ell_{0.75}$ vs. $B: (25-x,x)$

Notes: (b,r) denotes a sample with b blue (bad) signals and r red (good) signals. In comparative tasks, the subject chooses x, the minimal number of red balls needed from Box B for it to be preferred to Box A. In the belief-elicitation task, ℓ_q denotes a lottery that corresponds to an exogenous probability q of being golden.

3.5 Treatments

There is a distinct possibility that subjects use heuristic solely based on sample size and proportion. In such cases, we would expect them to be insensitive to changes in the explicitly stated information structure. I introduce three between-subject treatments to test this. Treatments differed *only* in the stated ball compositions conditional on box type, and hence in the information structure. All other features (interface, tasks, sample sizes, ordering, and incentives) were identical across treatments. Table 3 provides the full

composition, which describes the information structures. The treatment involves three information structures. Two were certain information structures, one symmetric and one asymmetric. One information structure was uncertain and symmetric, which is called correlated. In the correlated treatment, subjects are taught that the information structure includes a random component. Therefore, conditional on box type, the signal precision is itself uncertain. Subjects are randomly assigned to a treatment at the beginning of the experiment. The between-subject implementation avoids comparative anchoring across information structures. Subjects could at any point in the experiment click to see the implemented information structure.

Subjects were explicitly taught their treatment's information structure using worked examples; Figure 4 shows the instruction screen for the symmetric treatment.

Table 3: Information structures by treatment

Treatment	Golden box	Wooden box
Symmetric	7 red, 3 blue	7 blue, 3 red
Asymmetric	8 red, 2 blue	4 blue, 6 red
Correlated	4 red + 6 random	4 blue + 6 random

Notes: "Random" balls are independently red/blue with equal probability. Thus, in the correlated treatment, part of the sample is informative (the fixed-color balls), and part is pure noise, generating uncertainty about effective signal precision.

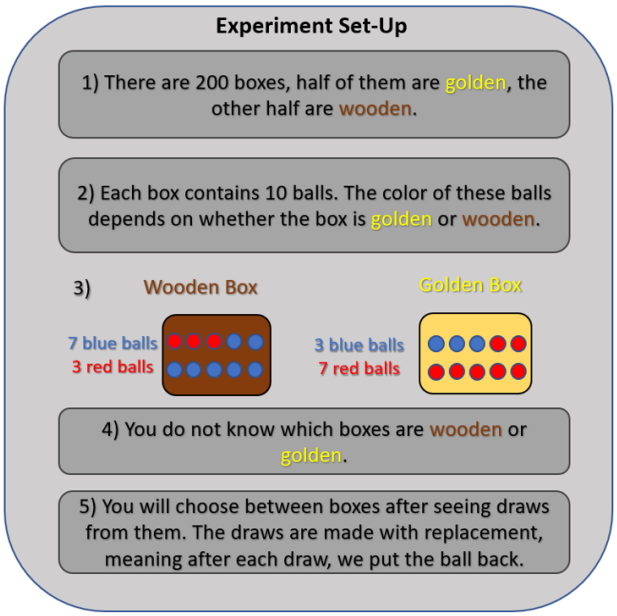


Figure 4: Treatment explanation screen (symmetric certain)

Notes: The asymmetric and correlated treatments are identical except that the ball composition is changed.

3.6 Confidence Elicitation

A central empirical prediction of Proposition 1 concerns the joint behavior of (i) sample-size neglect and (ii) choice confidence. To test this prediction, the experiment requires a confidence measure that is (a) tightly linked to the *specific choice* just made, (b) incentive-compatible with a wide range of non-EU and non-Bayesian models, and (c) minimally distortionary for the original choice task. Ideally, the measured confidence is also comparable between subjects. Standard self-reports do not satisfy these requirements, and classical incentive schemes often rely on strong assumptions that are likely violated here. I therefore implement a new confidence elicitation mechanism satisfying these conditions.

I define *confidence in a choice* as the DM's perceived certainty that the chosen option is the correct one (given their own subjective preferences). In formal models, this notion is naturally reflected in the *value of learning and revising*: if a subject is sufficiently confident that her current choice is correct, she should be unwilling to pay to (i) learn what the correct choice would be under the true data-generating process and (ii) obtain the opportunity to revise. Conversely, a strictly positive willingness to pay for such an opportunity is a revealed-preference manifestation of *lack of confidence*.

Immediately after each MPL choice, subjects face the following secondary decision as featured in Figure 5. This decision offers them two options that are relevant if this MPL decision is selected for payment. Either they retain their current MPL decision's switching point if it is selected for payment. Or they can choose a 50% chance of learning the optimal switching point and revising their decision, a 49% chance of using their current decision, and a 1% chance of receiving no bonus payment, which is the cost of learning.

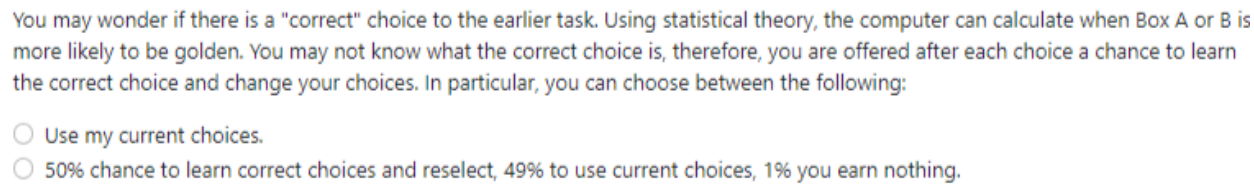


Figure 5: Confidence Elicitation Example

Notes: Subjects are taught how to use this interface and the choice implication. This interface appears at the bottom of the page where they make the original choices.

The conceptual contribution of the mechanism is to link confidence to the observable choice of learning the optimal choice. The key behavioral claim needed for the analysis is one-directional and minimal: *opting to learn is a sufficient condition for not having certainty in choice correctness, hence lacking confidence.* This implication is robust to standard departures from Bayesian updating and EU evaluation, provided the subject weakly prefers having

the option to correct mistakes when she anticipates potentially being wrong. This is therefore a lower bound and limits false positives. Additionally, the conservative and one-directional nature of the implication provides some safeguard against behavioral channels such as complexity aversion, regret, and randomization. These behavioral channels mostly push the subject to avoid learning information, even when they are not fully confident.[6] Furthermore, under full confidence, these channels are either ruled out or insufficient to generate learning.

To benchmark the incentivized confidence proxy against common practice, I also collected an unincentivized self-report of choice confidence at the end of the study.

3.7 Background.

I recruited 400 U.S.-based subjects on Prolific in September 2023 and implemented the study in oTree (Chen et al., 2016). Median completion time was 17 minutes. Subjects received a $2.50 show-up fee; in addition, approximately 60% earned a $5 performance-related bonus. Eligibility required a Prolific approval rate between 97%–99% and at least 100 prior completed studies. The experiment was preregistered; the preregistration is available at aspredicted.org/ui7g7.

4 Experimental Tests and Results

Overview. The theory delivers a sharp behavioral restriction: if a DM chooses as if knowing information structure, then her ICs in the sample space are parallel lines. This section turns that geometric implication into a set of transparent tests.

The empirical strategy has three ingredients. First, the comparative choice tasks yield switching thresholds, which are used to recover individual ICs. If ICs are parallel, then within-subjects, the slopes of different ICs should be identical. However, as the slope goes from 0 to ∞, it is more convenient to study the **standard deviation of IC angles** which goes from 0 to 90 degrees. Under information certainty, this dispersion should be near zero. Second, I compare the recovered ICs between the three treatments. This informs whether subjects mostly track sample characteristics or were sensitive to the explicitly stated information structure. Lastly, I consider the correlation between **sample-size neglect** and the choice to **opt to learn** (confidence). Information uncertainty predicts

[6] Non-instrumental value of information, such as curiosity, is also likely to be minimal and smaller for a confident subject (since they do not expect to learn).

a positive correlation, while heuristic use, confusion, and behavioral channel suggest no correlation or a negative correlation.

The next sections go into detail on how these measures are constructed and the exact tests. Table 4 in the appendix provides a summary of these measures.

Full and Sub-Sample Random error can mechanically generate apparent non-parallelism and attenuate treatment effects. Therefore, as pre-registered, I report results both in the full sample and in a pre-specified sub-sample that satisfies an explicitly taught monotonicity condition (equivalently, whose revealed ICs do not cross).[7] The sub-sample is therefore amenable to inattention or noisy behavior. There is only a 0.02% chance of having non-crossing ICs if a subject fully randomizes. Overall, 37% of subjects have 0 IC crossings, and they constitute this sub-sample. The theoretical maximum number of crossings is 8, and only 13% of subjects have more than 4 crossings. The appendix B.3 has further details about the monotonicity condition and measures of violation. Results are qualitatively robust to changes in the sub-sample threshold.

4.1 Parallelism of Indifference Curves

Revealed Indifference Curves. I first detail how ICs are recovered from the elicited threshold from the comparative choices. Denote by (b, r) a sample with b blue balls and r red balls. Recall that in a given choice, Box A displays a realized sample $(4-y, y)$ for $y \in \{0, 1, 2, 3, 4\}$. Box B will draw $N \in \{10, 25\}$ balls. The participant chooses the minimal $x \in \{0, \ldots, N\}$ such that Box B with $(N-x, x)$ is preferred to Box A with $(4-y, y)$. Denote this threshold by $x_{y,N}$. I elicit $x_{y,N}$ for each $y \in \{0, 1, 2, 3, 4\}$ and each $N \in \{10, 25\}$, yielding 10 thresholds per subject. In the comparative choice task, the threshold $x_{y,N}$ implies the IC through $(4-y, y)$ lies between $(N - x_{y,N}, x_{y,N})$ and $(N - (x_{y,N} - 1), x_{y,N} - 1)$. In the main analysis, I apply a midpoint rule and treat the IC as passing through

$$\left(N - \left(x_{y,N} - \tfrac{1}{2}\right),\ x_{y,N} - \tfrac{1}{2}\right),$$

yielding 10 revealed IC points from the comparative blocks. If the subject deems $(4-y, y)$ to be better or worse than $(N-x, x)$ for every x, then this midpoint does not exist. In this case, I take $(4-y, y)$'s IC to go through $(N, 0)$ if it is always better and $(0, N)$ if it is always worse.

[7] Within the sets of choices, to maintain consistency, subjects always start by evaluating the box with the lowest value, and each subsequent box is the next-highest in value. Therefore, it is straightforward to respect monotonicity as a subject only needs to remember their last choice.

Test of Parallelism. I plot the ICs of the three treatments on Figures 6 and 7. Figure 6 shows the full sample, and we observe that ICs gradually fan out. Figure 7 features the sub-sample; the fanning out is greater, suggesting a stronger violation that is not driven by noise or randomization. The parallelism of ICs fails at the aggregate level for both the full and sub-samples. To test parallelism, one ideally would measure and compare the slopes. However, the slope is hard to interpret as it goes from 0 to ∞. Instead, for every IC, I take its angle relative to the x-axis as a measure of its orientation, which goes from 0 to 90 degrees. I then take the standard deviation of all angles as a measure of parallelism. If ICs are parallel, then this SD should be close to 0.[8] The standard deviation is about 27 degrees, which differs sharply from the 0 degrees implied by parallel curves. This is largely independent of treatment, and holds for the full as well as the sub-sample. A full breakdown is given in Table 4 in the appendix. Since the aggregate ICs are not parallel straight lines, this suggests that information certainty does not account for aggregate behavior well.

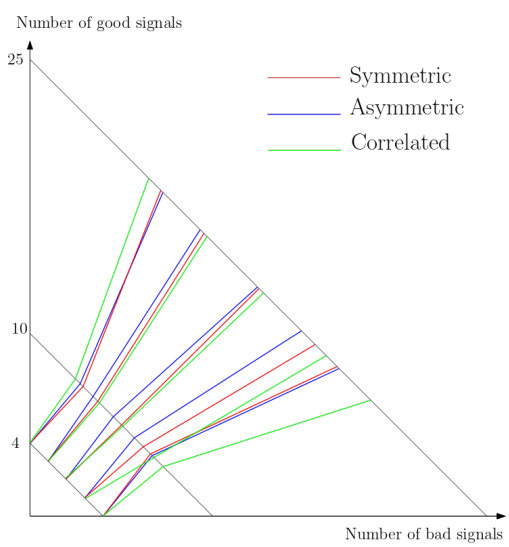


Figure 6: Full Sample IC

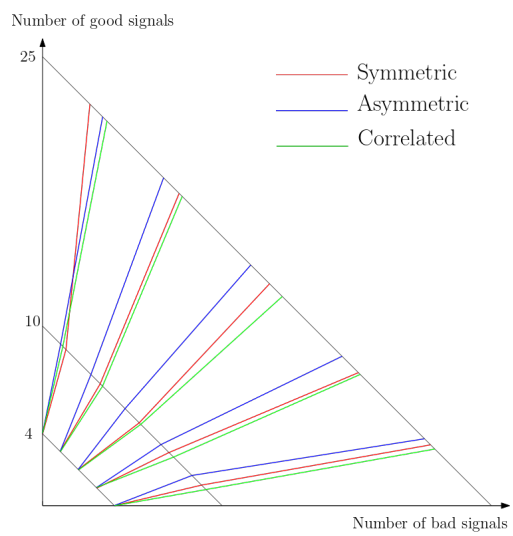


Figure 7: Sub-Sample IC

Notes: The x and y axes denote the number of blue (bad) and red (good) balls. Figures 6 and 7 denote the average ICs (assuming linearity) of the full sample and sub-sample, respectively.

Turning to the individual analysis, Figure 8 shows the distributions of standard deviations for the full and sub-sample samples. In the full sample, only 5% and 20% of subjects have ICs with standard deviations below 10 and 20 degrees, respectively. In the sub-sample, only 6% and 17% of subjects have ICs with standard deviations below 10 and 20 degrees, respectively. Therefore, separability is rejected at the individual level for almost all subjects. I note a spike at 33 degrees, which is the expected standard deviation

[8] It need not be perfectly zero due to the integer nature of samples.

should a subject fully randomize, see Table 4 for the random benchmark. However, this spike is not due to randomization but is due to subjects who fully neglect the sample size. Note further this spike's presence is even stronger for the sub-sample. The appendix B.5 further shows that sample-size neglect explains a large portion of the SD being non-zero.

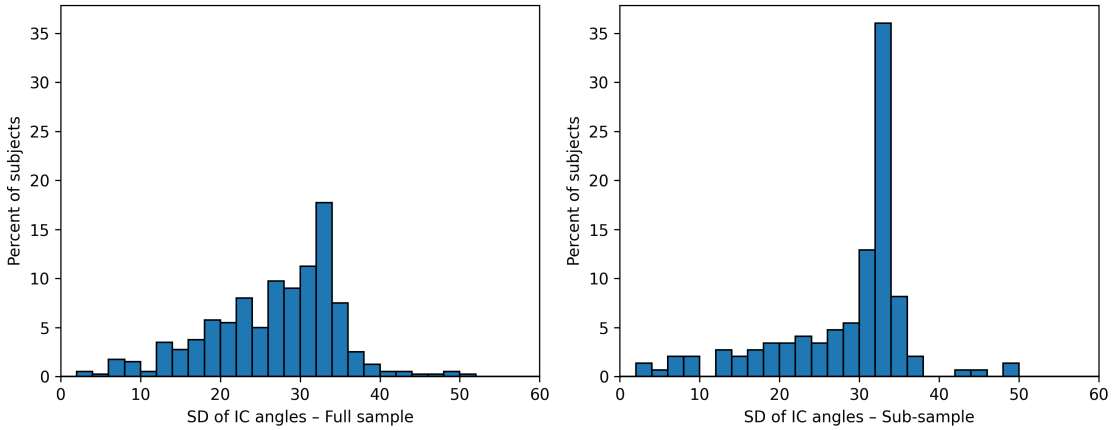


Figure 8: SD of IC Angles: Full Sample (left) and Sub-Sample (right)

I then test whether these subjects have ICs that gradually fan out. Recall that the thought experiment implies gradual fanning out. We know it suffices to have at least one IC that curves outward. Empirically, this is exemplified by the slope of an IC changing as the sample size increases. Additionally, to be consistent with fanning out, the slope needs to increase when the proportion of good signals is high and decrease when it is low. A subject is classified as consistent with the thought experiment if the slope of the IC through the sample with 4 good signals strictly increases or if the slope of the IC through the sample with 4 bad signals strictly decreases. I find that 76% and 61% of subjects have ICs that satisfy these conditions in the full and sub-sample, respectively. To benchmark these results, fix a subject's choice on tasks of sample size 10, and fully randomize their choice on those of sample size 25. Then 59% and 29% of subjects would satisfy these conditions for the full and sub-sample, respectively. This result is also largely independent of treatment; a full breakdown is given in Table 4 in the appendix.

Result 1. *Indifference curves are not parallel lines for 95% of subjects.*
Implication: *Subjects do not choose as if having information certainty.*

4.2 Treatment Sensitivity

Figures 6 and 7 suggest that subjects are essentially insensitive to the treatments. However, if subjects were sensitive to the likelihoods, then we would expect the ICs of the treatment to look different. Figure 9 shows the derived IC for a Bayesian EU agent. Note, the symmetric (red) and asymmetric (blue) both feature information certainty, and hence are parallel lines which differ only in slope; while the correlated treatment fan out as it features information uncertainty.

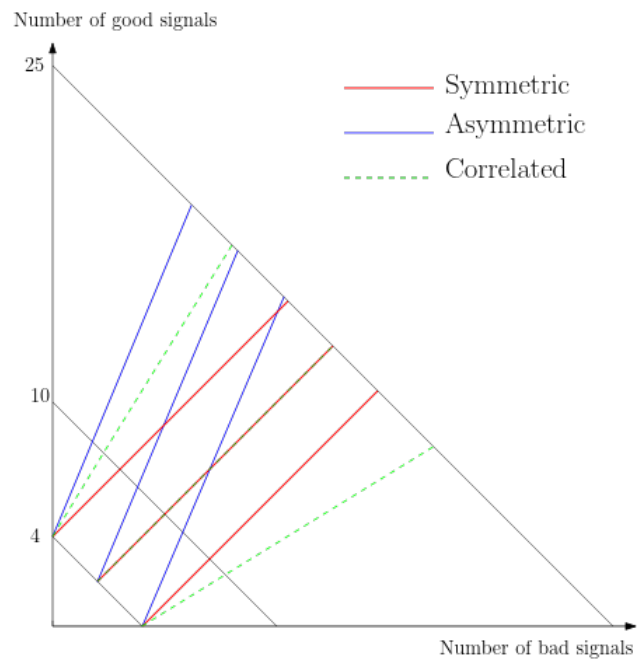


Figure 9: Bayesian EU IC

Notes: The x and y axes denote the number of blue (bad) and red (good) balls. The ICs obtained are calculated for a Bayesian EU subject.

To quantify the extent of the sensitivity to treatment, I estimate the following stacked regression pooling all ten choices of each subject:

$$\tilde{x}_{i,y,N} = \sum_{y',N'} \mathbb{1}_{\{(y,N)=(y',N')\}} \left[\alpha_{y',N'} + \beta^{\text{sym}}_{y',N'} \mathbb{1}\{t(i) = \text{sym}\} + \beta^{\text{asy}}_{y',N'} \mathbb{1}\{t(i) = \text{asy}\} + u_{i,y',N'}\right]. \quad (4)$$

Where $\mathbb{1}$ denotes the indicator function. Hence the stacked regression includes choice task specific fixed effect α. Let $t(i) \in \{\text{cor}, \text{sym}, \text{asy}\}$ denote subject i's treatment assignment, where cor (correlated) is the omitted category. For each $(y, N) \in \{0, 1, 2, 3, 4\} \times \{10, 25\}$, let $\tilde{x}_{i,y,N}$ be the inferred indifference point using the method detailed in Subsection 4.1. The above regression, therefore, tests whether the crossing points

on the $N \in \{10, 25\}$ and $y \in \{0, 1, 2, 3, 4\}$ ICs differ between treatments.

I test the joint null of $\beta^{\text{asy}}_{y,N} = \beta^{\text{sym}}_{y,N} = 0$, which would imply that there is no statistical difference between the three treatments for the particular y, N pair choice. For the full sample, for 8 out of 10 choices, I cannot reject the null for $p < 0.1$. For $(y, N) = (0, 10)$ and $(y, N) = (2, 10)$, I can reject them at $p = 0.093$ and $p = 0.023$ respectively. For the subsample, I cannot reject the null for 7 out of 10 choices. The full results are reported in Table 6 and 7 for the full and sub-sample, respectively. Overall, I find subjects largely choose solely according to sample proportion across the three treatments.

One might expect subjects to violate parallelism more when facing the correlated treatment with built-in information uncertainty. However, the treatment invariance suggests that the effect is very mild and that the explicitly stated information structure does not impact the violation of parallelism. I investigate whether the distributions of standard deviations differ by treatment. I cannot reject the null hypothesis for the full sample and sub-sample at any significance value $p \leq 0.10$ using the Kolmogorov-Smirnov test.

Result 2. *Choices are largely the same across the three treatment groups.*
Implication: *Subjects ignore the information structure and rely on sample characteristics.*

4.3 Confidence and Sample Size Neglect

Measurements. To assess whether sample-size neglect is correlated with confidence, two binary measures are derived from each choice. First, confidence is measured directly via opting to learn the correct choice. Second, sample-size neglect is displayed if the subject switches in the multiple-price list exactly when Box B's proportion of red balls is as close to Box A's proportion as possible (given the discreteness of sizes). Hence, this choice disregards the size difference fully. For each choice, the subject reveals the minimal $x_{y,N}$, $N \in \{10, 25\}$, such that $(N - x_{y,N}, x_{y,N}) \succsim (4 - y, y)$ for each $y \in \{0, 1, 2, 3, 4\}$. The subject's choice is consistent with a sample-size neglect if $|\frac{x_{y,N}}{N} - \frac{y}{4}| \leq 0.05$. Given the discreteness of the choices, this implies that the subject chose exactly to switch right after or right before the proportions equalize between the two samples. Figure 10 illustrates the type of ICs that would qualify. Note that the red band is thick due to the discreteness of sample sizes.

Overall, 39% and 61% of choices, for the full and sub-sample, respectively, displayed sample-size neglect. On the other hand, 25% and 19% of choices, for the full and sub-sample, respectively, had the subject opting to learn. Instead, if subjects are fully randomizing, then only 10% of choices would display sample-size neglect, and 50% of choices

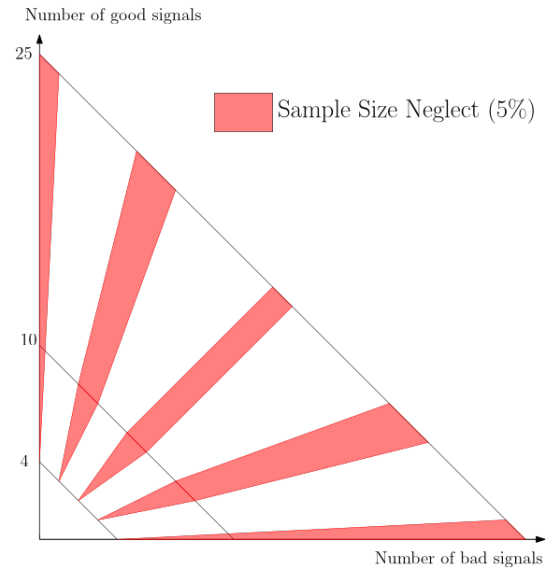


Figure 10: Sample Size Neglect ICs

Notes: The x and y axes denote the number of blue (bad) and red (good) balls. Figure 10 denotes ICs where the subject switches between Box A and B solely based on the sample proportion.

would display opting to learn. Around 40% of subjects never opted to learn, highlighting the earlier remark that this is a lower-bound measure of confidence. More details are in Appendix B.6 and Table 4.

Correlation Analysis. Figure 11 investigates the relationship between the two measures. In blue, I plot the proportion of choices for which a subject chooses to learn, conditional on the choice *not* displaying sample-size neglect. In red, the same is plotted for choices displaying sample-size neglect. If sample-size neglect was driven by confusion or complexity, it would be positively correlated with learning the optimal choice. However, blue bars are statistically higher than red bars (the error bars indicate 95% confidence intervals). The effect is stronger for the larger samples (25) and for the sub-sample: not displaying sample-size neglect implies a 2.4 times higher chance of opting to learn.

I also investigated this relationship using the following linear regression to test whether sample neglect is related to lack of confidence.

$$learn_{i,y,N} = \beta_0 + \beta_1 neglect_{i,y,N} + \lambda X_i + \epsilon_{i,y,N} \tag{5}$$

I denote choices by their (y, N) index and set $learn = 1$ if the subject opts to learn from that choice and 0 otherwise. Similarly, I set $neglect = 1$ if the choice exhibits sample-size neglect and 0 otherwise. Finally, X_i is a set of controls, including sex, ethnicity, time taken (in the whole study), age, and treatment dummies.

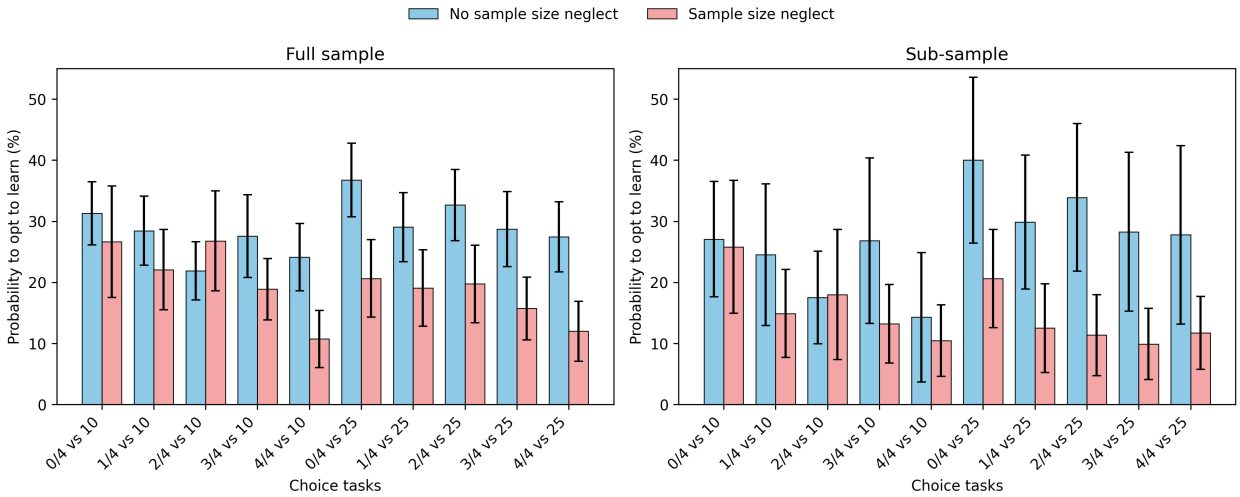


Figure 11: Full (Left) and Sub-Sample (Right) Confidence and Sample Size Neglect

Notes: Bars report the fraction of choices in which the subject opts to learn the statistically correct choice, plotted separately for choices classified as exhibiting sample-size neglect (red) versus not (blue). The 10 choices are displayed on the x-axis. Error bars indicate 95% confidence intervals.

Table 10 in the appendix presents the results for the regression as a linear probability model.[9] In both the full sample and the sub-sample, if the subject's choice displays sample-size neglect, then they are approximately 10% less likely to opt to learn. Notably, this is a large effect as the average probabilities of opting to learn are 25% and 19% for the full and sub-samples, respectively. I conclude that sample-size neglect is not due to a lack of confidence or noisy choices. On the contrary, a willingness to neglect the sample size and refer solely to sample proportions is associated with the subject being more confident. This finding aligns with the intuition from the thought experiment and the theoretical implications of having uncertainty in information.

Result 3. *Confidence is positively correlated to sample-size neglect.*

Implication: *Sample-size neglect does not stem from confusion. Instead, the pattern is consistent with information uncertainty.*

5 Discussion

Summary. This paper isolates and tests a modeling assumption that is usually taken for granted: that DMs know the information structure (likelihoods). Theorem 1 shows that, in a binary-quality environment, this assumption has sharp revealed-preference content

[9]Similar results are found for logit and probit models. As per my pre-registration, I also ran the regression with interaction terms and found similar results. See the online appendix for these additional regressions.

under mild monotonicity assumptions about the updating and choice rule. It is equivalent to separability and implies a linear evaluation of samples, hence *parallel* indifference curves (ICs) in sample space. This allows for a simple test of this assumption via the recovery of ICs via natural choices. Hence, the main contribution is to provide a general and complete identification condition of information certainty and to test it.

The main experimental takeaway is: subjects do not choose *as if* having information certainty. I also find that subjects are essentially insensitive to the explicitly given information structure. Furthermore, their choice tends to neglect the sample size in favor of the proportion of good signals. This pattern is also positively correlated with choice confidence.

I first discuss the implications of the results for experimental works and the plausibility of various interpretations of the current experimental findings. I then address the theoretical implications and discuss some additional concerns. Finally, I conclude by providing further details on the confidence elicitation mechanism, including demonstrating that it is well correlated with the standard unincentivized measure.

Experimental Discussion The experimental setting of the present paper is standard amongst belief-updating experiments that employ similar *ball-in-urn* type settings. In particular, the belief updating block is a standard belief elicitation task and replicates the key results about ICs: non-parallelism and insensitivity to treatments, see Figure 12. This makes it unlikely that the main findings are an artifact of the comparative elicitation format alone and likely have implications for standard belief updating experiments.

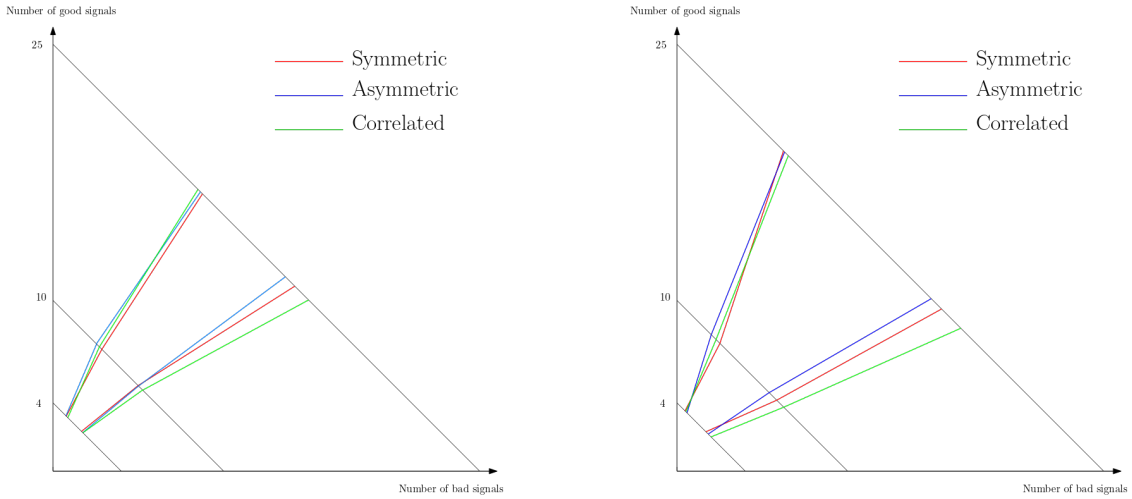


Figure 12: Full (Left) and Sub-Sample (Right) IC - Belief Updating

Notes: The x and y axes denote the number of blue (bad) and red (good) balls. Figure 11 denotes the average ICs (assuming linearity) obtained from the belief updating task of the full sample and sub-sample.

First, this literature typically assumes that the stated information structure is internally processed. Hence, a subject that deviates from the Bayesian posterior is taken to be non-Bayesian. However, given this present work's results, this statement must be qualified. Instead, one can only conclude that the subject is non-Bayesian *with respect to the given information structure*. This is somewhat understood, as experimentalists have appealed to subjective uncertainty, in an objective laboratory environment, to explain various biases (see Benjamin et al. (2016), Shmaya and Yariv (2016), and Augenblick et al. (2025)). The current work lends support to this channel as an alternative avenue by showing precisely that information certainty does fail, but also cautions against strong interpretations of the failure of Bayesian updating.

Second, this literature finds that subjects are responsive to the stated information structure. However, I find that subjects are almost fully insensitive to this stated information and rely solely on sample characteristics. What drives this gap? Before answering this question, it is useful to separately consider: (i) whether subjects are *aware* of the information structure, and (ii) whether they *use* it when mapping samples into choices. The design strongly supports (i) being true. Subjects received repeated explanations and visualizations of the treatment-specific ball compositions, passed comprehension checks that required understanding, and could at any point re-open the instruction screen displaying the information structure. The revealed-preference evidence supports the opposite for (ii): choices behave as if they place little weight on the treatment-specific likelihood parameters.[10]

The current paper's main goal was to cleanly test information certainty. The design and theory were therefore not meant to study why responsiveness did not differ by treatment and why $i)$ and $ii)$ are misaligned. Such a design would likely involve exploring the mental model, which is precisely orthogonal to the revealed-preference approach taken in the current work. However, the current paper does offer suggestive evidence for and against various potential interpretations and mental representations. First, we can rule out the following channels as main drivers:

1. **Noise, misunderstanding, randomization**: A randomizing or noisy subject could, in principle, display both ICs, which are non-parallel and treatment insensitive. However, subjects who fail comprehension checks are excluded. Further, if random responding were the primary driver, then restricting attention to a subsample exhibiting basic internal consistency should attenuate the patterns. Instead, the main qualitative facts, sample-size neglect and treatment invariance, are at least as

[10] I highlight that the treatment effect we do see is that subjects are more pessimistic towards bad news from the correlated treatment. This is consistent with Liang (2023)'s findings.

strong in the monotonicity-compliant subsample. To further illustrate why choice noise alone is insufficient, consider a Bayesian-EU agent who computes the Bayesian-optimal threshold $x_{y,N}$ but reports $x_{y,N}+\epsilon$ where $\epsilon \in \{-1, +1\}$ with equal probability. Such an agent generates an expected SD of IC angles of about 14 degrees, far below the observed mean (26.7 degrees), and moreover would have a 96.84% chance of violating monotonicity. This simple calibration suggests that the magnitude and structure of the deviations are not well explained by "Bayesian plus small random error."

2. **Cognitive uncertainty and complexity**: Subjects may resort to simple heuristics because the mapping from samples and likelihood parameters to posterior beliefs is cognitively demanding (e.g., Enke et al., 2024; Oprea, 2024). While these channels may play a role, two features are difficult to reconcile with a *pure* complexity/confusion account. First, the patterns strengthen in the monotonicity-compliant subsample, which includes subjects facing less complexity or cognitive limitations. Second, sample-size neglect is associated with *higher* revealed confidence (lower willingness to pay to learn and revise). Under weak assumptions that subjects weakly prefer correcting mistakes when they anticipate being wrong, a purely confusion-driven heuristic would tend to increase, not decrease, demand for correction. These findings, therefore, weigh against an interpretation in which sample-size neglect and treatment insensitivity are primarily driven by the difficulty of the task.

Instead, a parsimonious explanation for treatment insensitivity is that sample realizations (counts and proportions) are more **salient** than likelihood parameters, and that subjects rely on familiar cues from everyday environments (ratings, reviews, "more data feels more reliable"). On this view, subjects may ignore likelihoods not because they misunderstand them, but because the experimental representation makes sample statistics the dominant input into choice. This interpretation naturally aligns with evidence that probabilistic reasoning and updating depend strongly on presentation and framing (e.g., Gigerenzer and Hoffrage, 1995; Koehler, 1996). It also suggests a clear direction for future work: holding the underlying information structure fixed, vary whether information is presented as likelihood parameters versus pre-computed sample likelihoods/posteriors.

Alternatively, it could be that subjects subjectively use a different, potentially uncertain information structure that is stable across treatments. They might understand that red balls are good signals, but do not incorporate the treatment-specific parameter values. It could be because they do not encode them, don't trust them, or simply compress them into equivalent objects. Two features support this interpretation. In the thought experiment,

one is tempted to believe that sample reliability changes with sample size *even when given the information structure*. Additionally, such an interpretation predicts precisely the link between confidence and sample-size neglect. This explanation aligns with Benjamin et al. (2016) and Augenblick et al. (2025), which use information uncertainty to explain updating biases observed in experiments with information certainty.

Theoretical Discussion. Many theoretical results rely heavily on the assumption of information certainty. Therefore, the current paper is especially salient for environments where DMs observe multiple signals: social, dynamic, and repeated environments.[11] From limited works, we already know that various channels can behave very differently under information uncertainty (Huang, 2024; Aina, 2025; Chen, 2026). Additionally, the thought experiment and the current paper suggest that many important choice patterns cannot be rationalized by information certainty. Hence, while the literature has mostly relied on relaxing Bayesian updating or EU-preferences to accommodate certain patterns, an alternative possibility is the failure of information certainty.

The thought experiment also informs on models of uncertain information. The most popular models of updating under ambiguity are prior-by-prior and maximum likelihood updating. Both allow the DM to update on the uncertainty regarding the information structure. In these models, this updating depends only on the sample ratio, while the thought experiment can only be rationalized if it depended on both ratio and size. This suggests, at least for the case of updating maxmin EU beliefs, more sophisticated updating rules, such as Cheng (2022) and Tang (2025) might be more promising.

I now discuss two subtle theoretical results.

First, switching in the thought experiment—and the identification result—hinges on the **binary-quality assumption**, which is why the experiment must fix quality to be exogenously binary. This may be seen as a limitation, but this is a nuanced issue that reflects a fundamental non-identifiability issue. It turns out, choice data alone cannot disentangle *binarity of quality* from *information certainty*. Let C_{nb} be the set of choices rationalizable by Bayesian EU with *non-binary* quality and a known iid information structure, and let C_{unc} be the set of choicesrationalizable by Bayesian EU with *binary* quality but an iid information structure drawn from a (non-degenerate) mixture. Appendix A.3 shows $C_{nb} = C_{unc}$, so the two hypotheses are observationally equivalent. The following example illustrates the

[11] Many theoretical works collapse all information into a single signal realization and hence may seem to remain defensible. However, this argument is only partial and does not apply to many works that focus on binary signals. In general, multiple observations, from certain or uncertain information structures, cannot be collapsed into a single observation from a binary structure.

intuition.

Example 1. Let σ denote the probability of a good signal. Alice consider binary qualities g/b with prior $1/2$ and utilities $u(g) = 1$, $u(b) = 0$. For each object, she faces an iid signal structure that is uninformative with probability $1/2$ ($\sigma = 0.5$ regardless of quality) and informative with probability $1/2$ ($\sigma = 0.7$ in state g and $\sigma = 0.3$ in state b). Bob instead considers three qualities h, m, ℓ with utilities $1, 1/2, 0$ and priors $1/4, 1/2, 1/4$, and he is certain signals are iid with $\sigma \in \{0.7, 0.5, 0.3\}$ in states h, m, ℓ respectively. Both models induce the same prior over (σ, expected utility):

$$(\sigma, u) = (0.7, 1) \text{ w.p. } 1/4, \quad (0.5, 1/2) \text{ w.p. } 1/2, \quad (0.3, 0) \text{ w.p. } 1/4.$$

Since any sample's likelihood depends only on σ, both agents have the same posterior over expected utility after every sample and therefore make identical choices.

Given this impossibility result, it is important to fix the quality exogenously as binary. Applying the same rigor with which I approach certainty of information, it is, of course, impossible to rule out that some subjects may have failed to perceive quality as binary despite being told explicitly. However, over 95% of subjects violate information certainty; it is difficult to believe they all failed to understand that quality was binary. Note finally that the thought experiment can be adapted in the multi-quality world in two ways. First, one could be asked to choose the object with the higher probability of having a quality above a certain threshold, effectively binarizing the quality set. Second, if one object's quality distribution is deemed to first-order stochastically dominate another, given their respective samples, then the choice cannot flip when samples are multiplied.

Naturally, one desires a characterization of C_{unc}, the set of choice behavior rationalizable by a Bayesian with a prior over information structures (holding quality to be binary). This seems to be rather difficult. One might find it interesting that, under mild conditions, the switching behavior can occur at most once. That is, $s_1 \succ s_2$ and $n_1 s_1 \prec n_1 s_2$ is possible, but then $(n_1 + n_2)s_1 \succ (n_1 + n_2)s_2$ is impossible.

A wary reader may wish to raise the following concern: in addition to information certainty, signals were also assumed to be **iid conditional on the quality**. However, in this sample environment, these conditions are equivalent. Since signals are observed simultaneously within a sample, the DM cannot differentiate sample likelihoods based on the sequence of realizations. Consequently, we can restrict attention to exchangeable models.[12] Therefore, de Finetti (1931)'s representation theorem implies that her belief must

[12]The same conclusion is implied if she is unwilling to pay to observe the sequence.

be a mixture of iid information structures. In other words, denote the set of information structures by Σ, it must be that $p(s|g) = \int_{\Sigma} \Pi_{k=1}^{K} \sigma_g^{s^k} dF(\boldsymbol{\sigma})$. The conditional probability of observing a sample given a quality is determined by a distribution F over the set of possible information structures Σ. Therefore, F is the source of uncertainty, and information certainty is equivalent to a degenerate F. This implies signals must be perceived as conditionally iid given the quality alone whenever there is information certainty.

Confidence Elicitation In the experiment, the confidence elicitation mechanism is binary to reduce the cognitive load; however, non-binary and more granular measures can be elicited. Additionally, choosing to learn does not always reveal the optimal choice - else the subject may lack proper incentives for their original decision. Furthermore, the cost of learning is intentionally minimal, as subjects, especially on Prolific, are highly averse to losing payment. Finally, and very importantly, to avoid contamination from learning and reduce cognitive load, learning occurs only at the end of the experiment, when the chosen payment decision is revealed.

I collect, at the end of the experiment, a binary self-reported confidence. A subject is asked whether they believe to have chosen correctly for the majority of choices. Figure 13 shows that self-report is well correlated with opt-in to learn and sample-size neglect in the expected direction. On average, I find a subject who always opts to learn is 25% to 45% less likely to self-report that they are confident in the full and sub-sample, respectively. Note that only 52% and 64% of subjects self-report being confident. Hence, these are strongly correlated. Regression results on the relationships and analysis of the effect of feedback are included in appendix B.7 and B.8.

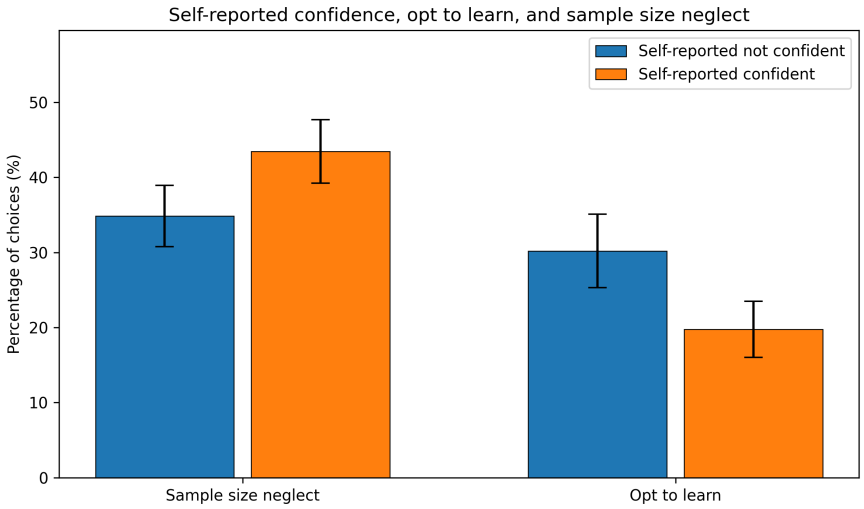


Figure 13: Full (Left) and Sub-Sample (Right) Confidence and Sample Size Neglect

Notes: "Opt to learn" is the percentage of the 10 comparative tasks in which a subject chose to learn the correction. "Sample-size neglect" is the percentage of the same tasks in which the subject switches when Box B's red-ball proportion is closest to Box A's proportion, i.e. $\left|\frac{x_{y,N}}{N} - \frac{y}{4}\right| \leq 0.05$. Self-reported confidence is confident if the subject claims to be confident that the majority of their choices were optimal. Bars show group means; whiskers denote 95% confidence intervals.

References

Aina, C. (2025). Tailored stories. Working Paper.

Aina, C. and F. Schneider (2026). Weighting competing models. Working Paper.

Augenblick, N., E. Lazarus, and M. Thaler (2025). Overinference from weak signals and underinference from strong signals. *The Quarterly Journal of Economics 140*(1), 335–401.

Bayes, T. and R. Price (1763). An essay towards solving a problem in the doctrine of chances. by the late rev. mr. bayes, f. r. s. communicated by mr. price, in a letter to john canton, a. m. f. r. s. *Philosophical Transactions (1683-1775) 53*, 370–418.

Benjamin, D. J., M. Rabin, and C. Raymond (2016). A model of nonbelief in the law of large numbers. *Journal of the European Economic Association 14*(2), 515–544.

Caticha, A. and A. Giffin (2006). Updating probabilities. In *AIP conference proceedings*, Volume 872, pp. 31–42.

Cerreia-Vioglio, S., D. Dillenberger, and P. Ortoleva (2015). Cautious expected utility and the certainty effect. *Econometrica 83*(2), 693–728.

Chambers, C. P. and N. S. Lambert (2021). Dynamic belief elicitation. *Econometrica 89*(1), 375–414.

Chan, K. (2025). An axiomatic model and test of grether (1980) and bayes' rule. *Available at SSRN 5084221*.

Chen, D. L., M. Schonger, and C. Wickens (2016). otree—an open-source platform for laboratory, online, and field experiments. *Journal of Behavioral and Experimental Finance 9*, 88–97.

Chen, J. Y. (2026). Sequential learning under informational ambiguity. *American Economic Review 116*(1), 209–245.

Cheng, X. (2022). Relative maximum likelihood updating of ambiguous beliefs. *Journal of Mathematical Economics 99*, 102587.

Coffman, K. B. (2014). Evidence on self-stereotyping and the contribution of ideas. *Quarterly Journal of Economics 129*(4), 1625–1660.

Cripps, M. W. (2018). Divisible updating. Working Paper.

Danz, D., L. Vesterlund, and A. J. Wilson (2022). Belief elicitation and behavioral incentive compatibility. *American Economic Review 112*(9), 2851–2883.

de Finetti, B. (1931). Funzione caratteristica di un fenomeno aleatorio. pp. 4:251–299.

Dominiak, A., M. Kovach, and G. Tserenjigmid (2023). Inertial updating. Working Paper.

Enke, B. and T. Graeber (2023). Cognitive uncertainty. *Quarterly Journal of Economics 138*(4), 2021–2067.

Enke, B., T. Graeber, R. Oprea, and J. Yang (2024). Behavioral attenuation. Working Paper.

Epstein, L. G. and Y. Halevy (2024). Hard-to-interpret signals. *Journal of European Economics Association 22(1)*, 393–427.

Epstein, L. G., J. Noor, and A. Sandroni (2010). Non-bayesian learning. *The BE Journal of Theoretical Economics 10*(1).

Gigerenzer, G. and U. Hoffrage (1995). How to improve bayesian reasoning without instruction: frequency formats. *Psychological review 102*(4), 684.

Gilboa, I. and D. Schmeidler (1993). Updating ambiguous beliefs. *Journal of economic theory 59*(1), 33–49.

Good, I. J. et al. (1963). Maximum entropy for hypothesis formulation, especially for multidimensional contingency tables. *The Annals of Mathematical Statistics 34*(3), 911–934.

Grether, D. M. (1980). Bayes rule as a descriptive model: The representativeness heuristic. *Quarterly Journal of Economics 95*(3), 537–557.

Griffin, D. and A. Tversky (1992). The weighing of evidence and the determinants of confidence. *Cognitive Psychology 24*(3), 411–435.

Halevy, Y., D. Walker-Jones, and L. Zrill (2023). Difficult decisions. Working Paper.

Hossain, T. and R. Okui (2013). The binarized scoring rule. *Review of Economic Studies 80*(3), 984–1001.

Huang, W. (2024). Learning about informativeness. *arXiv preprint arXiv:2406.05299*.

Jakobsen, A. M. (2023). Coarse bayesian updating. Working Paper.

Jaynes, E. T. (1957). Information theory and statistical mechanics. *Physical Review 106*(4), 620.

Karni, E. (2009). A mechanism for eliciting probabilities. *Econometrica 77*(2), 603–606.

Karni, E. (2018). A mechanism for eliciting second-order beliefs and the inclination to choose. *American Economic Journal: Microeconomics 10*(2), 275–285.

Karni, E. (2020). A mechanism for the elicitation of second-order belief and subjective information structure. *Economic Theory 69*(1), 217–232.

Koehler, J. J. (1996). The base rate fallacy reconsidered: Descriptive, normative, and methodological challenges. *Behavioral and brain sciences 19*(1), 1–17.

Kovach, M. (2021). Conservative updating. Working Paper.

Liang, Y. (2023). Learning from unknown information sources. *Management Science*. Accepted.

Möbius, M. M., M. Niederle, P. Niehaus, and T. S. Rosenblat (2022). Managing self-confidence: Theory and experimental evidence. *Management Science 68*(11), 7793–7817.

Nielsen, K. and L. Rigotti (2023). Revealed incomplete preferences. Working Paper.

Oprea, R. (2024). Decisions under risk are decisions under complexity. *American Economic Review 114*(12), 3789–3811.

Ortoleva, P. (2012). Modeling the change of paradigm: Non-bayesian reactions to unexpected news. *American Economic Review 102*(6), 2410–2436.

Pęski, M. and C. Stewart (2025). Non-distortionary belief elicitation. Working Paper.

Pires, C. (2002). A rule for updating ambiguous beliefs. *Theory and Decision 53*(2), 137–152.

Quiggin, J. (1982). A theory of anticipated utility. *Journal of economic behavior & organization 3*(4), 323–343.

Rabin, M. and J. L. Schrag (1999). First impressions matter: A model of confirmatory bias. *Quarterly Journal of Economics 114*(1), 37–82.

Shepherdson, J. C. (1980). Utility theory based on rational probabilities. *Journal of Mathematical Economics 7*(1), 91–113.

Shishkin, D. and P. Ortoleva (2023). Ambiguous information and dilation: An experiment. *Journal of Economic Theory 208*, 105610.

Shmaya, E. and L. Yariv (2016). Experiments on decisions under uncertainty: A theoretical framework. *American Economic Review 106*(7), 1775–1801.

Shore, J. and R. Johnson (1980). Axiomatic derivation of the principle of maximum entropy and the principle of minimum cross-entropy. *IEEE Transactions on Information Theory 26*(1), 26–37.

Tang, R. (2025). A theory of contraction updating. *Working Paper*.

Williams, P. M. (1980). Bayesian conditionalisation and the principle of minimum information. *British Journal for the Philosophy of Science 31*(2), 131–144.

Zhu, J., N. Chen, and E. P. Xing (2014). Bayesian inference with posterior regularization and applications to infinite latent svms. *Journal of Machine Learning Research 15*(1), 1799–1847.

Appendix

A Theoretical Appendix

A.1 Theorem 1

Throughout, I denote samples by x, y, z instead of s_1, s_2, s_3 to save a subscript. I denote by π_k^x the percentage of signals of type k in x and N_x the total sample size of x.

I show first that a linear representation is equivalent to the LR representation. Pick any set of values $(\sigma_{g,k}, \sigma_{b,k})_{k\in K}$. Pick any sample x and y. The LRs of the samples are

$$L(x) = \left[\prod_{k=1}^{K} \sigma_{g,k}^{\pi_k^x} / \prod_{k=1}^{K} \sigma_{b,k}^{\pi_k^x}\right]^{N_x} \text{ and } L(y) = \left[\prod_{k=1}^{K} \sigma_{g,k}^{\pi_k^y} / \prod_{k=1}^{K} \sigma_{b,k}^{\pi_k^y}\right]^{N_y}.$$

The DM always chooses the one with a higher LR. Therefore, we can take the $\log$ to obtain

$$x \succsim y \text{ if and only if } N_x \sum_{k=1}^{K} \pi_k^x \log(\frac{\sigma_{g,k}}{\sigma_{b,k}}) \geq N_y \sum_{k=1}^{K} \pi_k^y \log(\frac{\sigma_{g,k}}{\sigma_{b,k}}).$$

Then choosing $u_k = \log(\frac{\sigma_{g,k}}{\sigma_{b,k}})$ shows that the LR representation has a linear representation.

To show linear representation implies LR, suppose $\succsim$ is such that $\exists\{u_k\}$ such that

$$x \succsim y \text{ if and only if } N_x \sum_{k=1}^{K} \pi_k^x u_k \geq N_y \sum_{k=1}^{K} \pi_k^y u_k.$$

Then from above, suffice to find any set of values $(\sigma_{g,k}, \sigma_{b,k})_{k\in K}$ such $\sigma_{g,k} = \sigma_{b,k} \exp(\alpha u_k)$. While satisfying some constraints on the likelihoods

$$\forall k, \sigma_{b,k} \exp(\alpha u_k) \in (0,1) \text{ and } \sigma_{b,k} \in (0,1).$$

Note that this is simply a matter of scaling, as $\exp(\alpha u_k)$ is always positive. Not,e we can scale the utilities by α as it preserves $\succsim$. So we can always find a set of $\sigma_{b,k}$s small enough.

We show now that the axioms are equivalent to a linear representation. We extend $\succsim$ to the set of entries with rational numbers and apply Shepherdson (1980)'s result on multiplier spaces.

We proceed first by extending to the set of rational observations. To do so, we first show a version of the mixture independence axiom, which is implied by transitivity and separability. I start by defining a coarse type of mixture.

Definition 2. *For $\alpha \in [0,1]$, samples s_1 and s_2, if $\alpha s_1 \in \mathbb{N}_0^K$ and $(1-\alpha)s_2 \in \mathbb{N}_0^K$, then the mixture $s_1 \alpha s_2 = \alpha s_1 + (1-\alpha)s_2$.*

Therefore $s_1 \alpha s_2$ denotes the sample that is obtained by adding α proportion of s_1 to $(1-\alpha)$ proportion of s_2. Because samples are vectors with natural numbers as entries, I restrict this definition to whenever both proportions are themselves samples. Note that although I define a mixture operation, this is not a mixture space. Given this definition, I define the mixture independence axiom.

Axiom 3 (Mixture Independence)**.** *For all s_1, s_2, s_3 and α, $s_1 \succ (\sim) s_2 \Leftrightarrow s_1 \alpha s_3 \succ (\sim) s_2 \alpha s_3$ whenever $\alpha s_1, \alpha s_2, (1-\alpha)s_3 \in \mathbb{N}_0^K$.*

I show separability and transitivity implies mixture independence. Suppose we have $x \succsim y$, then we want to show $x\alpha z \succsim y\alpha z$. Note firstly that separability implies that $x \succsim y$ and $x' \succsim y'$ then $x + x' \succsim y + y'$. This is done by three applications of separability plus transitivity of $\succsim$. For mixture to be well defined, we have $\alpha x, \alpha y$ are samples of form $(\alpha x_1, ..., \alpha x_k)$ and $(\alpha y_1, ..., \alpha y_k)$ with integer entries. Note then that the smallest $\alpha = \frac{1}{N}$, which can work for both to be well defined, is when N is the greatest common divisor of x_ks and y_ks. Similarly, any α that can work is of the form $\frac{n}{N}$. Note then suffice to show that $x \succsim y$ implies $\alpha x \succsim \alpha y$, then using separability with $(1-\alpha)z$ yields mixture. First note that $\frac{1}{N}x \succsim \frac{1}{N}y$; if not, then we can apply separability and transitivity on both sides and obtain $x \prec y$. Then this gives for any $\frac{n}{N}$ we have $\frac{n}{N}x \succsim \frac{n}{N}y$ as desired. This completes the proof. The other direction is analogous as separability is also bi-directional.

Denote by $\mathbb{Q}$ the set of non-negative rationals. Then, $\mathbb{Q}^K$ is the set of samples with rational numbers of signals of each type. I define an extension of $\succsim$ on $\mathbb{Q}^K$, denoted by $\succsim^*$. Note that $\mathbb{Q}^K$ is what Shepherdson (1980) calls a *multiplier space* under mixtures with ratios in $\mathbb{Q}^K$. This is because, for any two rationals, their mixture by a ratio α that is itself a rational will be another rational. Shepherdson (1980) shows that $\succsim^*$ over such a space has a linear cardinal representation if and only if it satisfies three properties:

- Completeness+Transitivity.
- Closure of $\{\alpha \mid x\alpha y \succsim^* z\}$ and $\{\alpha \mid x\alpha y \precsim^* z\}$ in $\mathbb{Q}$ (as a subspace of $[0,1]$).
- Mixture Independence: $x \succsim^* y$ implies $x\alpha z \succsim^* y\alpha z$.

Therefore, if we can extend $\succsim$ to $\succsim^*$ while giving it these properties, then we know there is a linear representation for $\succsim$.

Pick any $x, y \in \mathbb{Q}^K$, then they can be rewritten as $(\frac{x_1}{d}, ...\frac{x_K}{d})$ and $(\frac{y_1}{d}, ..., \frac{y_K}{d})$ where $\tilde{x} = (x_1, .., x_K) \in \mathbb{N}_0^K$ and $\tilde{y} = (y_1, .., y_K) \in \mathbb{N}_0^K$. I extend $\succsim$ to $\succsim^*$ by letting $x \succsim^* y$ whenever $\tilde{x} \succsim \tilde{y}$. Note that there is more than one way to rewrite it $\tilde{x}$ and $\tilde{y}$, but by mixture, these must agree under $\succsim$. Suppose $x = \tilde{x} \cdot \frac{1}{d} = \overline{x} \cdot \frac{1}{c}$ and $y = \tilde{y} \cdot \frac{1}{d} = \overline{y} \cdot \frac{1}{c}$. Let $d > c$, then if $\overline{x} \succsim \overline{y}$, by mixture, $\overline{x}\frac{c}{d}0 \succsim \overline{y}\frac{c}{d}0$, which is equivalent to $\tilde{x} \succsim \tilde{y}$.

Note $\succsim^*$ is complete by definition. For any two vectors, x and y, with rational numbers as entries, it suffices to find the least common multiple of the denominators to obtain $\tilde{x}, \tilde{y} \in \mathbb{N}_0^K$.

Consider transitivity of triples, x, y, z samples. Then rewrite them as $x = \frac{\tilde{x}}{d}$, $y = \frac{\tilde{y}}{d}$, and $z = \frac{\tilde{z}}{d}$ where $\tilde{x}, \tilde{y}, \tilde{z} \in \mathbb{N}_0^K$. Then suppose $x \succsim^* y$, then $\tilde{x} \succsim \tilde{y}$ and similarly $\tilde{y} \succsim \tilde{z}$. So by transitivity of $\succsim$, $\tilde{x} \succsim \tilde{z}$ which implies $x \succsim^* z$.

Mixture is exactly like transitivity. Pick any x, y, α, z, we can rewrite all the terms as $\tilde{x}, \tilde{y}, \tilde{z}$. Then if $x \succsim^* y$, we have $\tilde{x} \succsim \tilde{y}$. Which implies $\tilde{x}\alpha\tilde{z} \succsim \tilde{y}\alpha\tilde{z}$, which implies $x\alpha z \succsim^* y\alpha z$.

Continuity. Note that $\{\alpha \mid x\alpha y \succsim^* z\}$ is the same as $\{\alpha \mid \forall\kappa, (\kappa x)\alpha(\kappa y) \succsim \kappa z\}$. Suppose that $x \succ y$, this implies that for $\alpha^+ > \alpha$, $(\kappa x)\alpha^+(\kappa y) \succ (\kappa x)\alpha(\kappa y)$. This monotonicity is guaranteed by mixture independence. Now we will show instead that $U = \{\alpha \mid \forall\kappa, (\kappa x)\alpha(\kappa y) \succ \kappa z\}$ is open. Let $\alpha^* \in U$, and suppose $\beta > \alpha^*$, then $\beta \in U$ by monotonicity. Now we want to find $\beta < \alpha^*$ such that $\beta \in U$. Suppose, $\exists\beta < \alpha^*$ such that $\beta \notin U$, then $(\kappa x)\alpha(\kappa y) \succ \kappa z \succ (\kappa x)\beta(\kappa y)$. But then by Archimedeaness, we know we can find a convex combination of the α and β mixtures such that it lies above z and yet below α.

A.2 Proposition 1

We assume that F is a probability measure with marginals F_g, F_b that each admit decompositions into finitely many atoms and absolutely continuous parts whose densities are continuous and strictly positive on finitely many intervals contained in $(0, 1)$. Formally, let $\omega \in \{g, b\}$ and let $\sigma \in (0, 1)$ be a likelihood, then the measure over information structure is of the form

$$F_\omega(d\sigma) = \sum_{j=1}^{J} \alpha_{\omega j}\delta_{x_j}(d\sigma) + \sum_{\ell=1}^{L_\omega} \mathbb{1}_{I_{\omega\ell}}(\sigma) f_{\omega\ell}(\sigma) d\sigma,$$

where $x_j \in (0, 1)$ are fixed points on which F_ω may assign non-zero probability and $\alpha_{\omega j} \geq 0$. Each interval $I_{\omega\ell} = [u_{\omega\ell}, v_{\omega\ell}] \subset [0, 1]$ is such that $f_{\omega\ell}$ is continuous and strictly

positive on $I_{\omega\ell}$. There are finitely many such intervals and points: $J < \infty$ and $L_\omega < \infty$.

Then let a sample s have ratio $r(s)$ of good signals. Denote by $h_r(\sigma) = r \log \sigma + (1 - r) \log(1-\sigma)$. Note that h always has a unique maximizer $\sigma = r$. Given (ω, σ), the likelihood of a sample $ns = (ns_g, ns_b)$ is

$$\Pr(ns|\omega, \sigma) = \binom{n(s_g + s_b)}{ns_g} \sigma^{ns_g}(1-\sigma)^{ns_b} = \binom{n(s_g + s_b)}{ns_g} e^{n(s_g+s_b)h_{r(s)}(\sigma)}.$$

Hence, the overall likelihood is

$$\Pr(ns|\omega) = \binom{n(s_g + s_b)}{ns_g} K_\omega(n; r(s)), \;\; K_\omega(n; r(s)) = \int_{(0,1)} e^{n(s_g+s_b)h_{r(s)}(\sigma)} F_\omega(d\sigma).$$

This then implies that the posterior, given prior p of being good, and after cancelling out of the combinatorial term, is

$$q_F(ns) := \Pr(\omega = g|ns) = \frac{pK_g(n; r(s))}{pK_g(n; r(s)) + (1-p)K_b(n; r(s))}.$$

We now apply Laplace's method to analyze the limit of the integral asymptotically. We know an atom $a_{\omega j}\delta_{x_j}$ contributes exactly $a_{\omega j} e^{n(s_g+s_b)h_{r(s)}(x_j)}$.

Let x^* be the maximizer of $h_{r(s)}$ on $[u, v]$. Then the interval term will contribute

$$\int_u^v f(\sigma) e^{n(s_g+s_b)h_{r(s)}(\sigma)} d\sigma = c(n(s_g + s_b))^{-\alpha} e^{n(s_g+s_b)h_{r(s)}(x^*)}(1 + o(1)),$$

with $c > 0$, where $\alpha = \frac{1}{2}$ if $r \in [u, v]$ and $\alpha = 1$ otherwise. Since F_ω has finitely many pieces, it follows that for each $\omega \in \{g, b\}$, there is a 1) maximum exponential rates $m_\omega(r)$ for the ratio r among the pieces of F_ω, 2) a smallest polynomial exponent $\alpha_\omega(r(s))$ and the sum $C_\omega(r)$ of the corresponding leading constants such that

$$K_\omega(n, r(s)) = C_\omega(r(s))(n(s_g + s_b))^{-\alpha_\omega(r(s))} e^{n(s_g+s_b)m_\omega(r(s))}(1 + o(1))$$

Then the posterior converges $q_F(ns) \to \psi(r)$ and crucially depends only on the ratio, where

$$\psi(r) = \begin{cases} 1, & m_g(r) > m_b(r) \text{ OR } m_g(r) = m_b(r) \text{ and } \alpha_g(r) < \alpha_b(r), \\ 0, & m_g(r) < m_b(r) \text{ OR } m_g(r) = m_b(r) \text{ and } \alpha_g(r) > \alpha_b(r), \\ \dfrac{p\, C_g(r)}{p\, C_g(r) + (1-p)\, C_b(r)}, & m_g(r) = m_b(r),\; \alpha_g(r) = \alpha_b(r). \end{cases}$$

Now we know that the Bayesian posterior favors samples with a higher ratio of good signals *conditional on equal sample size*. This and the fact that the posterior, in the limit, depends only on the ratio implies $\psi(r)$ is increasing in r. Let s_1, s_2 be two samples with ratios r_1, r_2. If $r_1 > r_2$, the heuristic chooses ns_1 as $n \to \infty$, so

$$U_H(ns_1, ns_2 \mid F) = q_F(ns_1) \to \psi(r_1) = \max\{\psi(r_1), \psi(r_2)\},$$

and similarly if $r_2 > r_1$. If $r_1 = r_2$, then the posterior implies both $q_F(ns_1)$ and $q_F(ns_2)$ converge to the same limit, so the heuristic payoff converges to that common value as well. Therefore

$$U_H(ns_1, ns_2 \mid F) \to \max\{\psi(r_1), \psi(r_2)\}.$$

Thus $U_B(ns_1, ns_2 \mid F)$ and $U_H(ns_1, ns_2 \mid F)$ converge to the same limit, which proves

$$\lim_{n\to\infty} \Big(U_B(ns_1, ns_2 \mid F) - U_H(ns_1, ns_2 \mid F) \Big) = 0.$$

A.3 Non-binary qualities with known accuracy is equivalent to binary quality with unknown accuracy

I show in this section that relaxing the assumption that the mental model has binary qualities, with known signal likelihoods, produces the same behavioral predictions as holding the binary state assumption and relaxing instead that signals likelihoods are known.

I say that $\succeq$ has a non-binary Bayesian expected utility representation with known accuracy if there is a set of qualities $q \in Q$, a positive and non-zero utility assigned to each quality $u(q)$, and for each quality a likelihood over signals of each type $\sigma_q = (\sigma_{q,1}, .., \sigma_{q,K})$, a prior $p(q)$ over qualities, such that the value of an object with sample x is:

$$\begin{aligned} V_{multi}(x) &= \sum_{Q} u(q)p(q|x) \\ &= \sum_{q} u(q)p(q)\frac{\prod_{k\in K} \sigma_{q,k}^{x_k}}{p(x)} \\ &= \sum_{q} u(q)p(q)\frac{\prod_{k\in K} \sigma_{q,k}^{x_k}}{\sum_{q'} p(q') \prod_{k\in K} \sigma_{q',k}^{x_k}}. \end{aligned}$$

I show that if $\succeq$ has the above representation, then it also has a binary quality representation with accuracy uncertainty with Bayesian updating and expected utility.

Then a Bayesian expected utility maximizer with a set $q \in Q$ of potential accuracies, distribution p_g, p_b over accuracies given quality, and $p(g)$ priors behave as follows. Note for each accuracy q, I denote the vector by $\sigma_q = (\sigma_{q,1,}, .., \sigma_{q,K})$. Note that $\frac{p(g)p_g(q)}{p(g)p_g(q)+p(b)p_b(q)} = p(g|q)$. First note now the DM chooses based on posterior; the value of an object associated with x is

$$\begin{aligned} V_{unc}(x) &= p(g|x) \\ &= \sum_q p(g,q|x) \\ &= \sum_q \frac{p(g,q)p(x|g,q)}{\sum_q [p(g,q') + p(b,q')]p(x|q')} \\ &= \sum_q p_g(q)p(g) \frac{\prod_{k \in K} \sigma_{q,k}^{x_k}}{\sum_{q'} [p_g(q')p(g) + p_b(q')p(b)] \prod_{k \in K} \sigma_{q',k}^{x_k}} \end{aligned}$$

To show the equivalence of the two representations, suffice to show that first for a given $u(q)$ and $p(q)$, we can find p_g, p_b and $p(g)$ such that $u(q)p(q) = p_g(q)p(g)$ and $p(q) = p(g)p_g(q) + p(b)p_b(q)$. And then show that for a given p_g, p_b and $p(g)$ we can find $u(q)$ and $p(q)$ where the equations hold.

Start with fixed $p(q)$ and $u(q)$, note that since the utility is linear, we can normalize $u(q)$ so that all $u(q) \in (0,1)$ and $\sum_q u(q)p(q) \in (0,1)$. Set $p(g) = \sum_{q'} u(q')p(q') \in (0,1)$. And set $p_b(q) = \frac{[1-u(q)]p(q)}{1-\sum_{q'} u(q')p(q')} \in (0,1)$. Then finally set $p_g(q) = \frac{u(q)p(q)}{\sum_{q'} u(q')p(q')}$. These guarantee the equations hold. Note these are always probabilities and sum up to 1.

B Experimental Appendix

B.1 Summary of Main Measures

Table 4: Summary Statistics of Measures

	Pooled		Symmetric		Asymmetric		Correlated		Rand. Bench.	
	Full	Sub	Full	Sub	Full	Sub	Full	Sub	Full	Sub
Standard Deviation of IC Angles	26.7	28.1	26.9	28.0	26.2	28.1	27.0	28.2	33.0	33.0
Thought Exp. Consistent (subjects)	76%	61%	77%	64%	92%	70%	72%	53%	59%	29%
Sample Size Neglect (choices)	39%	61%	37%	59%	38%	57%	43%	64%	10%	10%
Opt to Learn (choices)	25%	19%	22%	19%	25%	16%	27%	22%	50%	50%
N	400	147	140	44	128	43	132	60	400	147

Notes: The sub-sample are made of subjects satisfying the monotonicity condition. The random benchmark derives the standard deviation, opting to learn and sample-size neglect, assuming the subject fully randomizes. For the measure of thought experiment consistency, I fix a subject's choice on the 4 vs 10 comparative task and randomize their 10 vs 25 choice.

B.2 Descriptive Statistics

Table 5: Descriptive statistics of controls by treatment and monotonicity sub-sample (complete cases only)

	Pooled		Symmetric		Asymmetric		Correlated	
	Full	Sub	Full	Sub	Full	Sub	Full	Sub
Female (%)	38.4	34.5	36.1	32.6	41.9	31.0	37.3	38.6
Age	39.5 (13.9)	37.8 (13.1)	40.3 (14.5)	37.3 (13.2)	40.4 (14.3)	36.7 (13.3)	37.8 (12.8)	39.0 (13.1)
Time taken (min)	21.2 (12.4)	20.8 (11.1)	21.7 (11.6)	22.1 (11.6)	19.8 (11.0)	20.4 (9.8)	22.1 (14.3)	20.1 (11.8)
Ethnicity: White (%)	63.4	66.2	61.7	65.1	68.5	71.4	60.3	63.2
Ethnicity: Black (%)	14.1	9.2	12.8	2.3	11.3	7.1	18.3	15.8
Ethnicity: Asian (%)	12.0	12.0	12.0	11.6	9.7	7.1	14.3	15.8
Ethnicity: Mixed (%)	6.5	7.0	6.8	9.3	8.9	11.9	4.0	1.8
Ethnicity: Other (%)	3.9	5.6	6.8	11.6	1.6	2.4	3.2	3.5
N	383	142	133	43	124	42	126	57

Notes: The table restricts to complete cases for sex, ethnicity, age, and `timetaken`.

B.3 Monotonicity Condition

Recall that red balls are good signals in every treatment. This monotonicity implies that $(x, n) \succsim (x-1, n)$, which in turn implies $x_{y,N} \geq x_{y-1,N}$. A subject violates monotonicity if $x_{y-1,N} > x_{y,N}$ for any of the comparative tasks. There are 8 such comparisons, and the figure below plots the distribution of violation numbers. In the comprehension task, subjects are explicitly taught the monotonicity condition concerning red balls. Subjects can only start the actual tasks after demonstrating they understand the monotonicity condition.

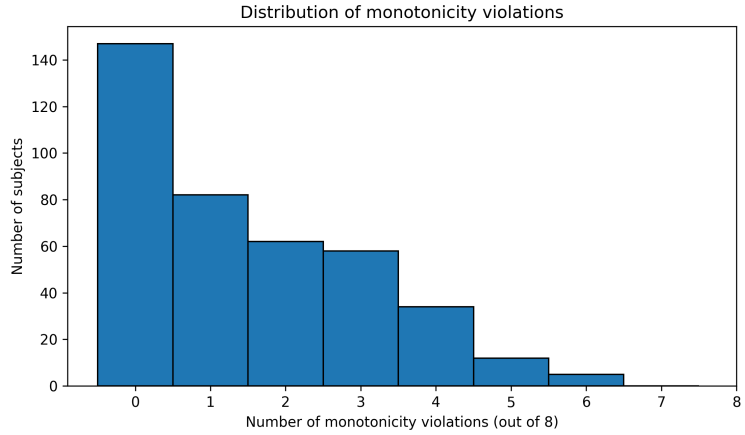


Figure 14: Distribution of the number of monotonicity violations (out of 8)

B.4 Treatment Sensitivity

Table 6: Treatment effects on inferred indifference midpoints

Point (N, y)	β^{asy} (SE)	β^{sym} (SE)	p_{joint}
$(10, 0)$	0.590 (0.361)	0.714 (0.349)	0.093
$(10, 1)$	0.170 (0.277)	−0.247 (0.262)	0.261
$(10, 2)$	0.514 (0.209)	0.039 (0.216)	0.023
$(10, 3)$	0.322 (0.245)	0.122 (0.263)	0.415
$(10, 4)$	−0.261 (0.350)	−0.418 (0.361)	0.500
$(25, 0)$	1.661 (0.965)	1.716 (0.926)	0.111
$(25, 1)$	1.180 (0.719)	0.568 (0.662)	0.260
$(25, 2)$	0.500 (0.590)	0.436 (0.569)	0.645
$(25, 3)$	0.508 (0.685)	0.304 (0.686)	0.756
$(25, 4)$	−0.660 (0.920)	−0.596 (0.925)	0.728

Notes: The regression includes a full set of point fixed effects (one indicator for each (N, y)). The omitted treatment category is the correlated treatment; β_{sym} and β_{asy} therefore measure the difference in $x_{i,y,N}$ relative to the correlated treatment for the symmetric and asymmetric treatments, respectively. Standard errors (in parentheses) are clustered at the individual level. The column p_{joint} reports the p-value of the joint Wald test of $H_0 : \beta_{\text{sym}} = \beta_{\text{asy}} = 0$.

Table 7: Treatment effects on inferred indifference midpoints (Sub-sample)

Point (N, y)	β^{asy} (SE)	β^{sym} (SE)	p_{joint}
$(10, 0)$	0.771 (0.425)	0.164 (0.372)	0.188
$(10, 1)$	0.703 (0.301)	0.239 (0.313)	0.064
$(10, 2)$	0.790 (0.244)	0.042 (0.315)	< 0.001
$(10, 3)$	0.542 (0.300)	0.082 (0.373)	0.121
$(10, 4)$	0.103 (0.389)	−0.204 (0.477)	0.784
$(25, 0)$	0.712 (1.188)	0.318 (1.100)	0.834
$(25, 1)$	0.969 (0.913)	0.110 (0.921)	0.536
$(25, 2)$	1.647 (0.660)	0.489 (0.819)	0.029
$(25, 3)$	1.053 (0.700)	0.208 (0.953)	0.263
$(25, 4)$	0.119 (0.969)	−0.248 (1.173)	0.946

Notes: This is the same as Table 6 but restricted to the subsample.

B.5 Sample Size Neglect and Non-Parallel ICs

I regress the standard deviation of angles, SD_i, on the proportion of times, out of the 10 choices, a subject displays sample-size neglect, P_i. Additionally, X_i is a set of controls, including sex, ethnicity, time taken (in the whole study), age, and treatment dummies. I estimate regression (6), with results presented in Table 8

$$SD_i = \beta_0 + \beta_1 P_i + \lambda X_i + \epsilon_i.$$

Table 8: Non-Parallel ICs and Sample Size Neglect

	SD of Angles of ICs - SD_i			
	(1)	(2)	(3)	(4)
P_i	11.6	11.7	19.9	19.7
	(0.94)	(1.00)	(1.8)	(1.9)
Controls/Treat.Dummy	No	Yes	No	Yes
Full/Sub-Sample	Full	Full	Sub	Sub
R^2	0.19	0.20	0.47	0.48
N	400	379	147	140

Note: Robust standard errors in brackets.

A subject who always displays sample-size neglect has an SD of 12 (20) degrees higher than a subject who never displays sample-size neglect for the full (sub) sample.

B.6 Distribution of Opt-in to Learn and Sample Size Neglect

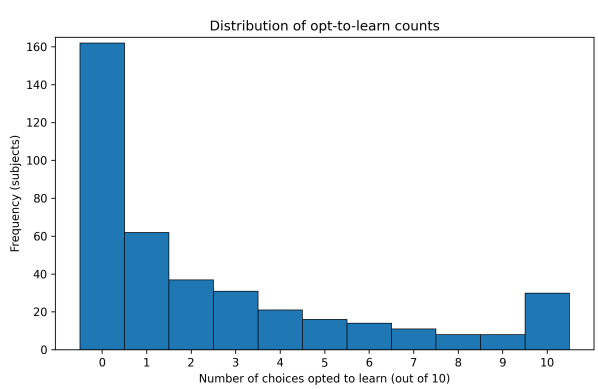


Figure 15: Dist. of opt-to-learn

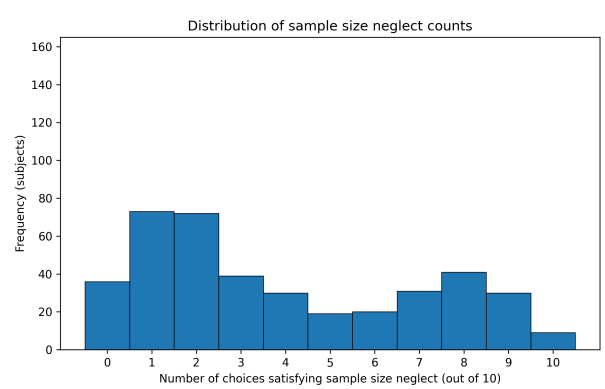


Figure 16: Dist. of sample-size neglect

Figure 17: **Opt-to-learn and sample-size neglect across the 10 comparative tasks.** *Notes*: Each panel plots, at the subject level, the number of (out of 10) comparative tasks in which the subject (left) opted to learn the correction or (right) satisfies the sample-size neglect criterion $\left|\frac{x_{y,N}}{N} - \frac{y}{4}\right| \leq 0.05$.

B.7 Confidence Measures Correlation

Denote by O_i the percentage of times (out of 10) that a subject i opts to learn, so the higher this is, the less confident a subject is. Let C_i represent the binary self-reported confidence measure, where $C_i = 1$ if the subject reports believing they made mostly correct choices and $C_i = 0$ otherwise. Finally, let X_i be a set of controls and treatment dummies.

I run the following regression (7) as a linear probability model. The regression results are in Table 9. On average, a subject who always opts to learn is 25% and 45% less likely to report that they are confident than someone who always opts out in the full sample and sub-sample, respectively. Note that only 52% and 64% of subjects self-report to be confident in the full and sub-samples, respectively. Hence, the effects are significant both in magnitude and in statistical significance, as Table 9 shows. See the online appendix for logit and probit results, which are consistent.

$$C_i = \beta_0 + \beta_1 O_i + \lambda X_i + \epsilon_i \tag{6}$$

Table 9: Self-Reported and Elicited Confidence

	Self-Reported Confidence - C_i			
	(1)	(2)	(3)	(4)
O_i	-0.26	-0.27	-0.47	-0.44
	(0.08)	(0.07)	(0.12)	(0.13)
Controls/Treat.Dummy	No	Yes	No	Yes
Full/Sub-Sample	Full	Full	Sub	Sub
N	400	379	147	140

Note: Robust standard errors in brackets.

B.8 Confidence Feedback Effect

I also consider how subjects adjusted their choices after receiving feedback. This part of the analysis was not pre-registered and is the only exploratory investigation in the paper. A total of 46 subjects were provided with the optimal choice and provided the opportunity to reselect. Nearly all subjects took the opportunity to change their choices. Amongst those who had not already chosen optimally (45 out of 46), only 7% (3 out of 45) did not change their choices.

Given the small sample size, I found some evidence that subjects' choices improved after feedback. In the sub-sample of subjects (12), the average absolute distance from the

optimal choice decreased from 4.41 before learning to 2.33 after learning, a statistically significant difference ($p \approx 0.04$, paired t-test). However, in the full sample, the average absolute distance only decreased from 5.17 to 4.67, which was not statistically significant.

B.9 Sample Size Neglect and Confidence

Table 10: Sample Size Neglect and Confidence

	Opting to learn $learn_{i,y,N}$			
	(1)	(2)	(3)	(4)
sample-size neglect, $neglect_{i,y,N}$	-0.10	-0.10	-0.12	-0.11
	(0.013)	(0.014)	(0.022)	(0.026)
Controls/Treat.Dummy	No	Yes	No	Yes
Full/Sub-Sample	Full	Full	Sub	Sub
N (number of choices)	4000	3860	1470	1410

Note: Robust standard errors in brackets. The full sample consists of 400×10 choices, subjects with missing controls were excluded. Full descriptive statistics can be found in the appendix.

Online Appendix

A Theoretical Notes

A.1 Updating Rules Details

I first begin with a proposition that shows that the thought experiment conflicts with updating rules that are weakly monotone in the likelihood ratio. Then, I individually analyze the above rules.

Proposition 2. *A preference relation $\succeq$ is said to display "switching" if $\exists s_1, s_2, \kappa$ such that $s_1 \succ s_2$ and $\kappa s_1 \prec \kappa s_2$. A preference relation $\succeq$ is said to be derived from an updating rule that is weakly monotonic in the likelihood ratio if $\exists \sigma_g, \sigma_b$ such that $L(s_1|\sigma_g, \sigma_b) \geq L(s_2|\sigma_g, \sigma_b)$ implies $s_1 \succeq s_2$. If $\succeq$ is derived from an updating rule that is weakly monotonic in the likelihood ratio, then it cannot display switching.*

Proof: Suppose $\succeq$ is derived from a rule that is weakly monotone in the likelihood ratio. Then $s_1 \succ s_2$ implies $L(s_1|\sigma_g, \sigma_b) > L(s_2|\sigma_g, \sigma_b)$. Then, if it displays switching, we must have some $s_1 \succ s_2$ and yet $\kappa s_2 \succ \kappa s_2$. So we must have $L(s_1|\sigma_g, \sigma_b)^\kappa < L(s_1|\sigma_g, \sigma_b)^\kappa$ which contradicts the earlier statement.

Table 11: Updating Rules Rejected by Thought Experiment and Relation to Theorem 1

Updating Rule[a]	Theorem 1	Indifference Curves
Bayesian updating (Bayes and Price, 1763)	Yes	Parallel Lines
Grether updating (Grether, 1980; Möbius et al., 2022)	Yes	Parallel Lines
Weighted Bayesian (Epstein et al., 2010; Kovach, 2021)	Yes	Parallel Lines
Divisible updating (Cripps, 2018)	Yes	Parallel Lines
Confirmatory bias (Rabin and Schrag, 1999)	Yes	Parallel Lines
Prior-by-prior updating (Pires, 2002)	No	Homothetic[c] Lines
Max. Lik. updating (Gilboa and Schmeidler, 1993)	No	Homothetic Lines
Size/proportion model (Griffin and Tversky, 1992)	No	Homothetic Curves
Coarse Bayesian (Jakobsen, 2023)	No	Thick Parallel Lines

[a] The online appendix goes over each of these rules in detail. The only updating rule, with information certainty, that can accommodate the thought experiment to the author's knowledge, does so by relaxing monotonicity with respect to the likelihood ratio, see Dominiak et al. (2023).
[c] ICs are "homothetic" if the slope is constant anywhere on a given ray through the origin. This means the shape of the IC is preserved through scalar multiplication.

I now provide some additional details for the updating rules of Table 11. I show that a wide class of models of non-Bayesian updating are functions of the likelihood ratio in this binary state setting. Denote by $\ell_g(s) = p(s|g, \sigma)$ where s is a sample. And denote by $p_B(g|s)$ the Bayesian posterior.

Bayesian updating: The Bayesian posterior ratio is proportional to the likelihood ratio and also, therefore, strictly increasing.

Grether updating: $p(g|s) = \frac{p(g)^\beta \ell_g(s)^\delta}{p(g)^\beta \ell_g(s)^\delta + (1-p(g))^\beta \ell_b(s)^\delta}$.

Note the posterior ratio of the states is:$[\frac{p(g)}{1-p(g)}]^\beta [\frac{\ell_g(s)}{\ell_b(s)}]^\delta$ where $\delta \geq 0$ is the signal reaction term. Therefore, the posterior ratio is weakly increasing and a function of the likelihood ratio whenever $\delta \geq 0$. If $\delta > 0$, which is the standard estimate, otherwise the DM is ignoring information, then it is strictly increasing.

Weighted Bayesian: $p(g|s) = \alpha p^* + (1-\alpha) p_B(g|s)$.

This updating rule is a convex combination of the Bayesian posterior and some arbitrary belief p^*. Taking p^* to be the prior would lead to underreaction. As the Bayesian posterior increases in the likelihood ratio, this updating rule is also. Similarly, whenever $\alpha < 1$, whenever the DM does update, it is strictly increasing.

Divisible Updating: It is shown in Cripps (2018) that a divisible updating rule must be homogeneous of degree 0 to the likelihoods of a signal. Therefore, if two samples have the same likelihood ratio, they will have the same posterior under a divisible updating rule. So, the updating rule is a function of the likelihood ratio.

Coarse Bayesian: this updating rule stipulates that there are convex subsets $P_1, .., P_N$ of $[0, 1]$ each with a "representative" probability $p_1 \in P_1, ..., p_n \in P_n$. The updating rule says that if $p_B(g|s) \in P_i$, then $p(g|s) = p_i$. So if the Bayesian posterior is in P_i, then the posterior is p_i. As convex subsets must be intervals, this updating rule is a function of the Bayesian posterior, which is a function of the likelihood ratio. This updating rule is not strictly increasing but weakly increasing and, therefore, cannot account for the thought experiment.

Size/Proportion Model: Griffin & Tversky propose, for a restrictive environment a regression that attempts to capture the weight of proportion of good signals, π, and sample size, N, in a DM's belief updating. Their regression only applies with prior $p = 0.5$ and when the signal structure is symmetric with $\sigma_g = 1 - \sigma_b$. So technically this is not an

updating rule. Their regression can be mapped as an updating rule in this specific setting. In particular, they estimate:

$$\ln(\ln(\frac{p(g|s)}{1-p(g|s)})) = \alpha_1 \ln(2\pi - 1) + \alpha_2 \ln(N) + \epsilon.$$

The idea here is that $\alpha_1 = \alpha_2$ implies Bayesian updating when the prior is uninformative $p(g) = 0.5$. Note that this is not separable but can still not accommodate the thought experiment. If two samples of size N_1, N_2 are multiplied to κN_1, κN_2, then they have a $+\alpha_2 \ln(\kappa)$, and therefore any inequalities are preserved.

Confirmatory Bias: This is technically a special type of perception rule; my setting is a little different as signals arrive together in one batch, whereas they model sequential observation with binary signals. However, the sequence turns out not to matter, so a faithful way of importing their model is to assume the DM has a bias for a state and may misperceive a signal for the other state as a signal for the biased state with probability q. Therefore, each sample (π, N) is changed to $(\pi(1-q), N)$ or $(\pi + (1-\pi)q, N)$, the rest is Bayesian updating.

Suppose a DM uses such an updating rule and perceives signals as iid. Then, by Theorem 1, the updating rule is strictly monotonic if and only if the $\succeq$ is separable. As we already have transitivity, completeness, and continuity.

Let $x = (\pi_1, N_1)$ and $y = (\pi_2, N_2)$. If $x \succeq y$ then the DM's belief σ_g, σ_b and bias in updating $q \in (0,1)$ satisfy

$$\frac{\sigma_g^{\pi_1(1-q)N_1}(1-\sigma_g)^{(1+\pi_1(q-1))N_1}}{\sigma_b^{\pi_1(1-q)N_1}(1-\sigma_b)^{(1+\pi_1(q-1))N_1}} \geq \frac{\sigma_g^{\pi_2(1-q)N_2}(1-\sigma_g)^{(1+\pi_1(q-1))N_2}}{\sigma_b^{\pi_2(1-q)N_2}(1-\sigma_b)^{(1+\pi_2(q-1))N_2}}.$$

This then implies that

$$\pi_1(1-q)N_1 \ln(\frac{\sigma_g}{\sigma_b}) + (1+\pi_1(q-1))N_1 \ln(\frac{1-\sigma_g}{1-\sigma_b}) \geq \pi_2(1-q)N_2 \ln(\frac{\sigma_g}{\sigma_b}) + (1+\pi_2(q-1))N_2 \ln(\frac{1-\sigma_g}{1-\sigma_b}).$$

Now let $z = (\pi_3, N_3)$ then note the new logged likelihoods of $x + z$ and $x + y$ are the following

$$\begin{aligned}&[\pi_1(1-q)N_1 + \pi_3(1-q)N_3] \ln(\frac{\sigma_g}{\sigma_b}) + [(1+\pi_1(q-1))N_1 + (1+\pi_3(q-1))N_3] \ln(\frac{1-\sigma_g}{1-\sigma_b}) \\ &\quad \geq [\pi_2(1-q)N_2 + \pi_3(1-q)N_3] \ln(\frac{\sigma_g}{\sigma_b}) + [(1+\pi_2(q-1))N_2 + (1+\pi_3(q-1))N_3] \ln(\frac{1-\sigma_g}{1-\sigma_b}).\end{aligned}$$

Note separability holds, and therefore, the updating rule is a monotonic function of the likelihood ratio.

Prior-by-Prior Updating: In prior-by-prior updating, the agent believes there is a set of information structures, updates with respect to all of them, and uses the worst posterior to evaluate the object. Then it is straightforward to see that if some information structure (σ_g, σ_b) provides the worst posterior for some signal s, then it will still provide the worst posterior for signal $k \times s$. This is by the likelihood ratio formulation. But then, $s_1 \succ s_2$ will immediately imply $k \times s_1 \succ k \times s_2$.

Maximum Likelihood Updating: In maximum likelihood updating, the agent also believes there is a set of information structures. She instead always picks the posterior from the information structure that has the highest likelihood of generating the sample. By a similar logic, the information structure most likely to generate a sample is independent of scalar multiplication of the sample. This yields the result.

Inertial Updating: Dominiak et al. (2023) follow a long line of literature which tries to model updating via a minimization problem involving the prior, likelihoods and posterior Jaynes (1957); Good et al. (1963); Williams (1980); Shore and Johnson (1980); Caticha and Giffin (2006); Zhu et al. (2014). While the literature traditionally focused on Bayesian updating, Dominiak et al. (2023) contribute by showing it can be used to study non-Bayesian updating, and more importantly give it behavioral foundations. Their updating rule can be rewritten as

$$p(g|s) = \frac{g(p(g))f(\ell_g(s))}{g(p(g))f(\ell_g(s)) + g(1-p(g))f(\ell_b(s))}.$$

As f and g have flexible functional forms, the posterior need not be increasing in the likelihood ratio. The generality of this representation is due to the author's commitment to a simple axiomatization, and the paper offers several special cases that satisfy monotonicity in likelihood ratio. In private conversation, the authors have shared that they also strongly agree that monotonicity is a weak property.

A.2 General Confidence Elicitation and Incentive-Compatibility

I consider a subject who must choose from a set of actions $a \in A$. The subject has a payoff function $\pi : A \times A \to Z$. The set Z denotes the potential consequences of her choices. It could be objective, e.g., monetary values, or subjective, e.g., subjective belief about the probability of winning, and I assume there is an outcome z_w that is understood

by all subjects to be the worst one. $\pi(a, a^*)$ denotes the consequence should the subject choose a when a^* is the correct choice. Correct can be objective, as in the case of belief updating, or it could be subjective as in the case of dictator games or lottery choices. Finally, I denote by $s \in S$ a set of signal realizations. The subject may believe signals are correlated to the correct action a^*. I assume that $\pi(a, a^*)$ is uniquely maximized at $a = a^*$ for each a^*. This implies that not knowing the correct choice is payoff-relevant. I note that these signals do not provide any value of information regarding uncertainties intrinsic to the experiment (such as lottery outcomes). The only instrumental value they can provide is in terms of the correctness of action ex-ante any resolution of uncertainty. The subject has some belief about the correct choice a^*. I say a subject is confident in knowing a^* whenever they assign probability 1 to some $a^* \in A$. If a subject is confident, then nothing can change her belief about a^*. Therefore, a confident subject should assign zero instrumental value to any signal, whether the subject believes it to be correlated with the correct action or not. The following example illustrates one common experimental setting that this framework nests.

Example 2. Consider eliciting a subject's probabilistic belief p that an event E occurred via some incentive-compatible mechanism, Karni (2009); Hossain and Okui (2013). The correct belief, given the available information, is the Bayesian p^*. The subject reports p and is paid $\pi(p, p^*)$ that is uniquely maximized at $p = p^*$ whenever the elicitation is incentive-compatible. A set of signals could be to reveal to the subject the correct Bayesian posterior, in which case $S = [0, 1]$. This is only valuable if the subject lacks confidence in their report.

Given the above setup, I propose the following confidence elicitation mechanism:

- The subject is asked to submit an action $a \in A$ and a number $\delta \in [0, 1]$.
    1. With probability δ^2, they get z_w.
    2. With probability $1 - \delta$, they get $\pi(a, a^*)$.
    3. With probability $(1 - \delta)\delta$, they observe a signal s and can change their action.

In the case of the correct choice being objective and known to the researcher, she can set $S = A$ and allow the signal to reveal the correct action. The procedure, in this case, allows the subject to be paid as if they knew the correct action a^* with some probability. The subject's belief about a^* may be a probability distribution over A or a set of possible a^*s depending on the theory of confidence that is chosen. Any such theory generates values V_2 and V_3 for the second and third outcomes of the above mechanism. Furthermore, any

such theory can generate $V_2 < V_3$ only when the belief about a^* is not degenerate, and the signal is expected to be informative. I assume the DM's attitude towards the lottery generated by the mechanism satisfies FOSD. That is given any choice of δ, the DM faces a lottery with values $0, V_2$ and, V_3, normalizing the value of z_w to 0. Then FOSD implies that the DM picks $\delta > 0$ only if $V_2 > V_3$ which I show is only possible if she lacks confidence.

Proposition 3. *Suppose the DM's attitude regarding the lottery induced by the mechanism satisfies strict FOSD then $\delta > 0$ only if the DM lacks confidence.*

Proof: I will show that this holds for theories that assign either a probability over actions (over their correctness) and for theories that consider a set of actions to be correct. Suffice to show that $V_2 = V_3$ whenever the DM is confident in a^* being correct with probability 1. In this case, consider S^* as the set of signals the DM expects to be possible given a^*. No conventional theory of updating can assign a positive probability to a state that previously had 0 probability[13]. Therefore, the value of observing any signal is 0, as the DM does not expect to change her belief. And if her attitude towards the mechanism is strictly FOSD then she does not acquire information.

B Additional Regressions

B.1 Regression with Interaction Terms

I regress specifications (1) and (2) with treatment interaction as

$$SD_i = \beta_0 + \sum_{t \in T} \delta_t \beta_{1,k} P_i + \sum_{t \in T} \delta_t D_t + \lambda X_i + \epsilon_i, \text{ and}$$
$$o_d = \beta_0 + \sum_{t \in T} \delta_t \beta_{1,k} p_d + \sum_{t \in T} \delta_t D_t + \lambda X_i + \epsilon_d.$$

[13]The only exceptions are theories that consider observing signals which are not possible given the DM current belief such as Ortoleva (2012). However, the DM does not expect to receive such signals and hence in her information acquisition decision she does not account for these.

Table 12: Non-Parallel ICs and Sample Size Neglect

	SD of Angles of ICs - SD_i			
	(1)	(2)	(3)	(4)
$\beta_{1,sym}$	12.5	12.0	19.9	21.0
	(1.2)	(1.8)	(1.8)	(2.5)
$\beta_{1,asy}$	10.7	11.3	19.9	17.4
	(1.2)	(1.7)	(1.8)	(3.9)
$\beta_{1,cor}$	11.6	12.3	19.9	20.9
	(1.0)	(1.6)	(1.8)	(3.8)
Controls/Treat.Dummy	No	Yes	No	Yes
Full/Sub-Sample	Full	Full	Sub	Sub
R^2	0.19	0.22	0.47	0.49
N	400	386	147	141

Note: Robust standard errors in brackets.

Table 13: Opting to Learn and Sample Size Neglect

	Opt To Learn - o_d			
	(1)	(2)	(3)	(4)
$\beta_{1,sym}$	-0.13	-0.10	-0.12	-0.08
	(0.02)	(0.02)	(0.03)	(0.04)
$\beta_{1,asy}$	-0.09	-0.08	-0.16	-0.11
	(0.02)	(0.02)	(0.03)	(0.04)
$\beta_{1,cor}$	-0.09	-0.12	-0.10	-0.13
	(0.2)	(0.02)	(0.03)	(0.04)
Controls/Treat.Dummy	No	Yes	No	Yes
Full/Sub-Sample	Full	Full	Sub	Sub

Note: Robust standard errors in brackets.

B.2 Logit and Probit Regressions for Specification (2) and (3)

Table 14: Self-Report and Elicited Confidence - Logit

	Self-Reported Confidence - C_i			
	(1)	(2)	(3)	(4)
O_i	-1.11	-1.26	-1.91	-1.95
	(0.34)	(0.35)	(0.60)	(0.62)
Controls/Treat.Dummy	No	Yes	No	Yes
Full/Sub-Sample	Full	Full	Sub	Sub
N	400	391	147	146

Note: Robust standard errors in brackets.

Table 15: Self-Report and Elicited Confidence - Probit

	Self-Reported Confidence - C_i			
	(1)	(2)	(3)	(4)
O_i	-0.69	-0.79	-1.18	-1.20
	(0.21)	(0.21)	(0.37)	(0.38)
Controls/Treat.Dummy	No	Yes	No	Yes
Full/Sub-Sample	Full	Full	Sub	Sub
N	400	386	147	141

Note: Robust standard errors in brackets.

Table 16: Sample Size Neglect and Confidence- Logit

	Opting to learn o_d			
	(1)	(2)	(3)	(4)
p_d, sample size neglect	-0.53	-0.57	-0.73	-0.71
	(0.08)	(0.14)	(0.08)	(0.14)
Controls/Treat.Dummy	No	Yes	No	Yes
Full/Sub-Sample	Full	Full	Sub	Sub
N	4000	3860	1470	1410

Note: Robust standard errors in brackets.

Table 17: Sample Size Neglect and Confidence - Probit

	Opting to learn o_d			
	(1)	(2)	(3)	(4)
p_d, sample size neglect	-0.31	-0.33	-0.41	-0.41
	(0.05)	(0.05)	(0.08)	(0.08)
Controls/Treat.Dummy	No	Yes	No	Yes
Full/Sub-Sample	Full	Full	Sub	Sub
N	4000	3860	1470	1410

Note: Robust standard errors in brackets.

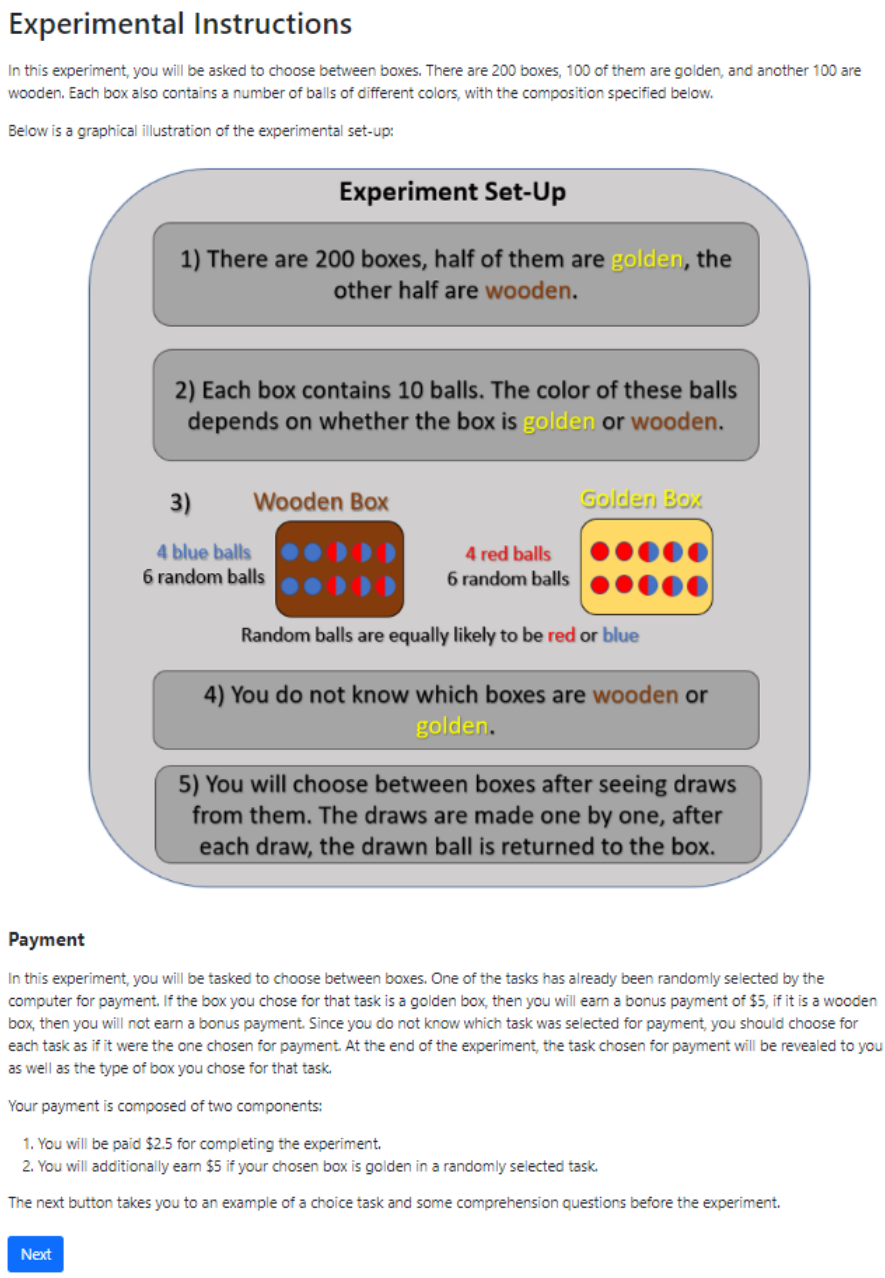
C Additional Table for Confidence and Sample Size Neglect

Table 18: Opting to Learn Given Sample Size Neglect

			% of Subjects Who Opt to Learn			
	% of Sample Size Neglect		Sample Size Neglect		No Sample Size Neglect	
Choice Task	Full	Subsample	Full	Subsample	Full	Subsample
0/4 vs 10	23%	42%	27%	26%	31%	27%
1/4 vs 10	39%	64%	22%	15%	28%	25%
2/4 vs 10	28%	34%	27%	18%	22%	18%
3/4 vs 10	58%	72%	19%	13%	28%	27%
4/4 vs 10	42%	71%	11%	10%	24%	14%
0/4 vs 25	39%	66%	21%	21%	37%	40%
1/4 vs 25	38%	54%	19%	13%	29%	30%
2/4 vs 25	38%	68%	20%	11%	33%	34%
3/4 vs 25	48%	69%	16%	10%	29%	28%
4/4 vs 25	42%	76%	12%	11%	27%	28%

D Experimental Instructions

D.1 Instructions



Experimental Instructions

In this experiment, you will be asked to choose between boxes. There are 200 boxes, 100 of them are golden, and another 100 are wooden. Each box also contains a number of balls of different colors, with the composition specified below.

Below is a graphical illustration of the experimental set-up:

Payment

In this experiment, you will be tasked to choose between boxes. One of the tasks has already been randomly selected by the computer for payment. If the box you chose for that task is a golden box, then you will earn a bonus payment of $5, if it is a wooden box, then you will not earn a bonus payment. Since you do not know which task was selected for payment, you should choose for each task as if it were the one chosen for payment. At the end of the experiment, the task chosen for payment will be revealed to you as well as the type of box you chose for that task.

Your payment is composed of two components:

1. You will be paid $2.5 for completing the experiment.
2. You will additionally earn $5 if your chosen box is golden in a randomly selected task.

The next button takes you to an example of a choice task and some comprehension questions before the experiment.

Next

Figure 18: Instructions: Correlated Treatment Instructions

The symmetric and asymmetric treatments were identical except balls were fully colored in either red or blue with the respective distributions.

Introduction

Hover to see the experimental set-up.

Throughout the experiment, you will have to choose between boxes. In this example, you are presented with two boxes. Each of the two boxes were randomly selected from the pool of 200 boxes. If the box you choose is golden and this choice is randomly selected by the computer, then you will receive an additional $5. Here is how boxes A and B differ:

1. The computer drew 5 balls from Box A one at a time, returning each ball to the box after it was drawn. Of the 5 balls drawn, 3 were red (and the other 2 were blue).
2. The computer will draw 10 balls from Box B and your choice can be based on the outcome of the draws.

Note that the more red balls drawn from a box, the more likely it is golden rather than wooden. So if you pick Box B when 6 red balls out 10 balls are drawn from it, you should still pick it for 7, 8, 9 and 10 red balls out of 10. The interface below is implemented to automatically make these selections for you.

The interface will automatically fill out your choices in the following way. If you click on the right column for a row, say you choose Box B when 4 red balls out of 10 are drawn from it , then for all the rows above this row the computer selects Box A and for all the rows below it selects Box B. Similarly, if you choose Box A when Box B draws 7 red balls out of 10 and click on the left column for that row, then for all the rows above, the computer selects Box A and for all the rows below it selects Box B. You can also refresh the page to start over anytime.

Choose Box A		Choose Box B
Box A - **3 out of 5** balls drawn were red.	○ ○	Box B - if **0 out of 10** balls drawn are red.
Box A - **3 out of 5** balls drawn were red.	○ ○	Box B - if **1 out of 10** balls drawn are red.
Box A - **3 out of 5** balls drawn were red.	○ ○	Box B - if **2 out of 10** balls drawn are red.
Box A - **3 out of 5** balls drawn were red.	○ ○	Box B - if **3 out of 10** balls drawn are red.
Box A - **3 out of 5** balls drawn were red.	○ ○	Box B - if **4 out of 10** balls drawn are red.
Box A - **3 out of 5** balls drawn were red.	○ ○	Box B - if **5 out of 10** balls drawn are red.
Box A - **3 out of 5** balls drawn were red.	○ ○	Box B - if **6 out of 10** balls drawn are red.
Box A - **3 out of 5** balls drawn were red.	○ ○	Box B - if **7 out of 10** balls drawn are red.
Box A - **3 out of 5** balls drawn were red.	○ ○	Box B - if **8 out of 10** balls drawn are red.
Box A - **3 out of 5** balls drawn were red.	○ ○	Box B - if **9 out of 10** balls drawn are red.
Box A - **3 out of 5** balls drawn were red.	○ ○	Box B - if **10 out of 10** balls drawn are red.

In order to proceed, please refresh the page and make the hypothetical selections (underlined) of the previous paragraph in the stated order to see for yourself how the interface works. The button will be enabled after you perform the selections of the above paragraph.

Next

Figure 19: Instructions: Payment and Choice Example

Introduction

Hover to see the experimental set-up.

Your choice was Box B whenever at least 8 out of 10 balls drawn from it are red.

You may wonder if there is a "correct" choice to the earlier task. Using statistical theory, the computer can calculate when Box A or B is more likely to be golden. You may not know what the correct choice is, therefore, you are offered after each choice a chance to learn the correct choice and change your choices. In particular, you can choose between the following:

- ○ Use my current choices.
- ○ 50% chance to learn correct choices and reselect, 49% to use current choices, 1% you earn nothing.

In the actual experiment, you will choose one of these two options for each task. **At the end of the experiment**, it will be revealed to you the task that was selected for payment and the correct choices. You will also be given a chance to change your choices depending on the option selected.

For example, you will learn the following:

The choice that gives the highest probability of picking a golden box is to pick Box B whenever at least 6 red balls are drawn from it.

Please select one of the two options to proceed.

Next

Figure 20: Instructions: Confidence Elicitation

Introduction

Hover to see the experimental set-up.

To sum up the process of the experiment:

1. You are tasked to choose between boxes, there are 16 tasks.
2. For each task, you can choose to potentially learn the correct choice.
3. After finishing all the tasks, you will be informed of the chosen task for bonus payment.
4. You will be given the correct choice and a chance to change your choice according to your earlier selection.
5. The experiment is then over, you will be informed of your bonus payment and given the completion code.

Before beginning the experiment, please answer a few comprehension questions. **You have three attempts**, if you answer incorrectly three times or more, please return the study.

How many boxes are there total?

Suppose you are choosing between Box A and B which are randomly chosen from the 200 boxes, if Box A is golden, does it mean Box B is wooden?

Suppose we draw 10 balls from Box A and B, returning the ball to the box after each draw. Box A has 7 red balls out of 10 and Box B has 6 red balls out of 10. Which one is more likely to be golden?

Next

Figure 21: Instructions: Comprehension Check

D.2 Choice Examples

Choice

Hover to see the experimental set-up.

You are offered a choice between two boxes, A and B. Each of these boxes were randomly picked from the 200 boxes. You will be paid $5 if the box you pick is golden and the computer has selected this task for payment. Box A and B were randomly picked the following way:

1. Box A was picked as follows:
    1. The computer drew 4 balls from each of the 200 boxes. They were drawn one at a time, returning each ball to the box after it was drawn.
    2. 52 out 200 boxes had 2 out 4 red balls drawn from them (2 other balls were blue).
    3. Box A was randomly selected from these 52 boxes.

2. Box B was picked as follows:
    1. The computer randomly picked Box B from the 200 boxes.
    2. The computer will draw 10 balls from Box B, one at at time and returning the drawn ball to the box.
    3. You can make your choice based on the number of red balls that are drawn from B.

Choose Box A		Choose Box B
Box A - **2 out of 4 (50%)** balls drawn were red.	○ ○	Box B - if **0 out of 10 (0%)** balls drawn are red.
Box A - **2 out of 4 (50%)** balls drawn were red.	○ ○	Box B - if **1 out of 10 (10%)** balls drawn are red.
Box A - **2 out of 4 (50%)** balls drawn were red.	○ ○	Box B - if **2 out of 10 (20%)** balls drawn are red.
Box A - **2 out of 4 (50%)** balls drawn were red.	○ ○	Box B - if **3 out of 10 (30%)** balls drawn are red.
Box A - **2 out of 4 (50%)** balls drawn were red.	○ ○	Box B - if **4 out of 10 (40%)** balls drawn are red.
Box A - **2 out of 4 (50%)** balls drawn were red.	○ ○	Box B - if **5 out of 10 (50%)** balls drawn are red.
Box A - **2 out of 4 (50%)** balls drawn were red.	○ ○	Box B - if **6 out of 10 (60%)** balls drawn are red.
Box A - **2 out of 4 (50%)** balls drawn were red.	○ ○	Box B - if **7 out of 10 (70%)** balls drawn are red.
Box A - **2 out of 4 (50%)** balls drawn were red.	○ ○	Box B - if **8 out of 10 (80%)** balls drawn are red.
Box A - **2 out of 4 (50%)** balls drawn were red.	○ ○	Box B - if **9 out of 10 (90%)** balls drawn are red.
Box A - **2 out of 4 (50%)** balls drawn were red.	○ ○	Box B - if **10 out of 10 (100%)** balls drawn are red.

You may wonder if there is a "correct" choice to this task. Using statistical theory, the computer can calculate when Box A or B is more likely to be golden. You may not know what the correct choice is, therefore, you are offered after each choice a chance to learn the correct choice and change your choices. In particular, you can choose between the following:

Choose an option:

Next

Figure 22: Comparative 4 vs 10 Balls Example

Choice

Hover to see the experimental set-up.

You are offered a choice between two boxes, A and B. Each of these boxes were randomly picked from the 200 boxes. You will be paid $5 if the box you pick is golden and the computer has selected this task for payment. Box A and B were randomly picked the following way:

1. Box A was picked as follows:
    1. The computer drew 4 balls from each of the 200 boxes. They were drawn one at a time, returning each ball to the box after it was drawn.
    2. 25 out 200 boxes had 0 out 4 red balls drawn from them (4 other balls were blue).
    3. Box A was randomly selected from these 25 boxes.

2. Box B was picked as follows:
    1. The computer randomly picked Box B from the 200 boxes.
    2. The computer will draw 25 balls from Box B, one at at time and returning the drawn ball to the box.
    3. You can make your choice based on the number of red balls that are drawn from B.

Choose Box A		Choose Box B
Box A - **0 out of 4 (0%)** balls drawn were red.	○ ○	Box B - if **0 out of 25 (0%)** balls drawn are red.
Box A - **0 out of 4 (0%)** balls drawn were red.	○ ○	Box B - if **1 out of 25 (4%)** balls drawn are red.
Box A - **0 out of 4 (0%)** balls drawn were red.	○ ○	Box B - if **2 out of 25 (8%)** balls drawn are red.
Box A - **0 out of 4 (0%)** balls drawn were red.	○ ○	Box B - if **3 out of 25 (12%)** balls drawn are red.
Box A - **0 out of 4 (0%)** balls drawn were red.	○ ○	Box B - if **4 out of 25 (16%)** balls drawn are red.
Box A - **0 out of 4 (0%)** balls drawn were red.	○ ○	Box B - if **5 out of 25 (20%)** balls drawn are red.
Box A - **0 out of 4 (0%)** balls drawn were red.	○ ○	Box B - if **6 out of 25 (24%)** balls drawn are red.
Box A - **0 out of 4 (0%)** balls drawn were red.	○ ○	Box B - if **7 out of 25 (28%)** balls drawn are red.
Box A - **0 out of 4 (0%)** balls drawn were red.	○ ○	Box B - if **8 out of 25 (32%)** balls drawn are red.
Box A - **0 out of 4 (0%)** balls drawn were red.	○ ○	Box B - if **9 out of 25 (36%)** balls drawn are red.
Box A - **0 out of 4 (0%)** balls drawn were red.	○ ○	Box B - if **10 out of 25 (40%)** balls drawn are red.
Box A - **0 out of 4 (0%)** balls drawn were red.	○ ○	Box B - if **11 out of 25 (44%)** balls drawn are red.
Box A - **0 out of 4 (0%)** balls drawn were red.	○ ○	Box B - if **12 out of 25 (48%)** balls drawn are red.
Box A - **0 out of 4 (0%)** balls drawn were red.	○ ○	Box B - if **13 out of 25 (52%)** balls drawn are red.
Box A - **0 out of 4 (0%)** balls drawn were red.	○ ○	Box B - if **14 out of 25 (56%)** balls drawn are red.
Box A - **0 out of 4 (0%)** balls drawn were red.	○ ○	Box B - if **15 out of 25 (60%)** balls drawn are red.
Box A - **0 out of 4 (0%)** balls drawn were red.	○ ○	Box B - if **16 out of 25 (64%)** balls drawn are red.
Box A - **0 out of 4 (0%)** balls drawn were red.	○ ○	Box B - if **17 out of 25 (68%)** balls drawn are red.
Box A - **0 out of 4 (0%)** balls drawn were red.	○ ○	Box B - if **18 out of 25 (72%)** balls drawn are red.
Box A - **0 out of 4 (0%)** balls drawn were red.	○ ○	Box B - if **19 out of 25 (76%)** balls drawn are red.
Box A - **0 out of 4 (0%)** balls drawn were red.	○ ○	Box B - if **20 out of 25 (80%)** balls drawn are red.
Box A - **0 out of 4 (0%)** balls drawn were red.	○ ○	Box B - if **21 out of 25 (84%)** balls drawn are red.
Box A - **0 out of 4 (0%)** balls drawn were red.	○ ○	Box B - if **22 out of 25 (88%)** balls drawn are red.
Box A - **0 out of 4 (0%)** balls drawn were red.	○ ○	Box B - if **23 out of 25 (92%)** balls drawn are red.
Box A - **0 out of 4 (0%)** balls drawn were red.	○ ○	Box B - if **24 out of 25 (96%)** balls drawn are red.
Box A - **0 out of 4 (0%)** balls drawn were red.	○ ○	Box B - if **25 out of 25 (100%)** balls drawn are red.

You may wonder if there is a "correct" choice to this task. Using statistical theory, the computer can calculate when Box A or B is more likely to be golden. You may not know what the correct choice is, therefore, you are offered after each choice a chance to learn the correct choice and change your choices. In particular, you can choose between the following:

Choose an option:

Next

Figure 23: Comparative 4 vs 25 Balls Example

Choice

Hover to see the experimental set-up.

You are offered a choice between two boxes, A and B. You will be paid $5 if the box you pick is golden and the computer has selected this task for payment. Box A and B are different in the following way:

1. Box A has a 25% chance of being golden.

2. Box B was randomly selected from the 200 boxes.
    1. The computer will draw 4 balls from Box B, one at at time and returning the drawn ball to the box.
    2. You can make your choice based on the number of red balls that are drawn from B.

Choose Box A			Choose Box B
Box A - **25%** chance of being golden.	○	○	Box B - if **0 out of 4 (0%)** balls drawn are red.
Box A - **25%** chance of being golden.	○	○	Box B - if **1 out of 4 (25%)** balls drawn are red.
Box A - **25%** chance of being golden.	○	○	Box B - if **2 out of 4 (50%)** balls drawn are red.
Box A - **25%** chance of being golden.	○	○	Box B - if **3 out of 4 (75%)** balls drawn are red.
Box A - **25%** chance of being golden.	○	○	Box B - if **4 out of 4 (100%)** balls drawn are red.

You may wonder if there is a "correct" choice to this task. Using statistical theory, the computer can calculate when Box A or B is more likely to be golden. You may not know what the correct choice is, therefore, you are offered after each choice a chance to learn the correct choice and change your choices. In particular, you can choose between the following:

Choose an option:

Next

Figure 24: Belief Elicitation 4 Balls Example

D.3 Pre-Payment and Payment

Thought Process

Before proceeding to showing you the task the computer selected for payment (and the correct choices), we'd like to know your thought process behind your choices. In particular, it would help us if you answered the following questions.

In the experiment, you were offered a chance to learn the correct choice. Sometimes you may have chosen to not use it. Which of the following best describes why you chose not to use it?

- ○ I did not use it because I knew the correct choice.
- ○ I did not use because of the 1% chance of not earning a bonus payment.
- ○ I always used it.

Overall, how sure are you that you made the right choices?

- ○ I know my choices were close to correct (for most tasks).
- ○ I do not know if my choices were close or not to correct (for most tasks).

In the experiment, how would you describe your choice when choosing between two boxes that both had balls drawn from them?

- ○ I relied mostly exclusively on the percentage of red balls drawn.
- ○ I used the percentage and weighted it by the total number of balls drawn.
- ○ I tried to estimate how likely each box was golden independently.
- ○ I used something else entirely.

Next

Figure 25: Unincentivized Confidence Elicitation

PrePayment

Hover to see the experimental set-up.

The task that was selected for payment was the following. We repeat the instructions below:

You are offered a choice between two boxes, A and B. Each of these boxes are randomly picked from the 200 boxes. You will be paid $10 if the box you chose is golden and if this choice is selected for payment. Box A and B are different in the following way:

1. We drew 4 balls from Box A, **2 out of 4 (50%)** are red.
2. We will draw 25 balls from Box B and your choice can be based on the outcome of the draws.

In this task, you chose box B if it drew at least **7 out of 25** red balls. And when given the chance to learn the optimal choice, you chose to have a 50% chance to learn the correct choices, 49% to use current choices, 1% you earn nothing. **You got the option to learn the correct choice and can reselect.**

Please enter the smallest number of red balls Box B needs to draw for you to choose it:

0

Below are the optimal choices for the different outcomes of Box A.

The draw from Box A	When to choose Box B
0 out 4 are red balls (the rest are blue).	If it draws at least 11 out of 25 red balls.
1 out 4 are red balls (the rest are blue).	If it draws at least 12 out of 25 red balls.
2 out 4 are red balls (the rest are blue).	If it draws at least 13 out of 25 red balls.
3 out 4 are red balls (the rest are blue).	If it draws at least 14 out of 25 red balls.
4 out 4 are red balls (the rest are blue).	If it draws at least 15 out of 25 red balls.

Next

Figure 26: Learning from the Confidence Elicitation Mechanism Example

Payment

You are done! Below we summarize your earnings:

In this task, you chose box B if it drew at least **7 out of 25** red balls. In the end, Box B drew **24 out of 25** red balls, which means you chose **Box B**. The box was **wooden** , you will not earn a bonus payment.

Additionally, the completion code for prolific is CKZ0D6QO.

If you have any questions, please reach us via Prolific.

Figure 27: Final Payment Example